

Performance Analysis

 Reattempt

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Detailed analysis of your performance in the AWS Certified Solutions Architect Associate - SAA C03 exam.

 Answer Distribution

0
Correct

0
Incorrect

64
Unattempted

 Domain Performance

Storage	0%
Databases	0%
Serverless	0%
Compute	0%
Monitoring & Management	0%
Security	0%
Networking	0%
Cost Optimization	0%

Final Score

**0/64**

Passing score: 48

Time Taken

**130m 0s**

Total time: 130 minutes



Question Analysis

✓ Correct	0
✗ Incorrect	0
⊖ Unattempted	64
🔖 Reviewed	0

Question-wise Analysis

Expand All

Collapse All

Question 1

Not Attempted

Storage



A company runs an application on several Amazon EC2 instances that store persistent data on an Amazon Elastic File System (Amazon EFS) file system. The company needs to replicate the data to another AWS Region by using an AWS-managed service solution in the MOST cost-effective manner. Which solution meets these requirements?

- ✗ Use the EFS-to-EFS replication to replicate the data to an EFS file system in another Region.

While EFS-to-EFS replication is a valid solution, AWS Backup is generally considered a more managed service and can offer better cost-effectiveness for long-term retention, backup policies and regional availability.

- ⊗ Run a nightly script to copy data from the EFS file system to an Amazon S3 bucket. Enable S3 Cross-Region Replication on the S3 bucket.

This solution requires managing your own scripts and is not a fully AWS-managed service. It also potentially involves higher operational overhead and costs compared to AWS Backup or EFS Replication.

- ⊗ Create a VPC in another Region. Establish a cross-Region VPC peer. Run a nightly rsync to copy data from the original Region to the new Region.

Similar to option B, this solution involves managing your own infrastructure and scripts. This is not an AWS-managed service and can incur significant operational overhead, complexity and cost.

- ✓ Use AWS Backup to create a backup plan with a rule that takes a daily backup and replicates it to another Region. Assign the EFS file system resource to the backup plan.

AWS Backup is a managed service specifically designed for backing up and replicating data across regions. It provides a cost-effective, policy-based approach with centralized management, making it the most suitable option.

Overall Explanation:

AWS Backup provides a fully managed, policy-based service for backing up and replicating data, satisfying the requirement for an AWS-managed service and cost-effective solution for replicating the EFS file system to another Region. While EFS-to-EFS replication can be considered, AWS Backup offers simplicity in policy management and a more comprehensive approach to data protection across services.

[View Discussions](#)[Ask AI](#)[Question 2](#)[Not Attempted](#)[Databases](#)

An e-commerce company is migrating its on-premises workload to the AWS Cloud. The workload currently consists of a web application and a backend Microsoft SQL database for storage. The company anticipates a surge in customer traffic during an upcoming promotional event. The new infrastructure in the AWS Cloud must be highly available and scalable, while minimizing administrative overhead. Which solution BEST meets these requirements?

- ⊗ Migrate the web application to two Amazon EC2 instances across two Availability Zones behind an Application Load Balancer. Migrate the database to Amazon RDS for Microsoft SQL Server with read replicas in both Availability Zones.

This option provides high availability for the application but using read replicas for the database is not the optimal way to achieve HA. Read replicas are typically used for read scaling and not for automatic failover in case of a primary database failure. It also increases administrative overhead compared to a managed solution.

- ⊗ Migrate the web application to an Amazon EC2 instance that runs in an Auto Scaling group across two Availability Zones behind an Application Load Balancer. Migrate the database to two EC2 instances across separate AWS Regions with database replication.

While this option offers scalability for the web application, replicating the database across AWS Regions introduces significant complexity and higher latency. Managing database replication on EC2 instances requires more administrative overhead than using a managed database service.

- ✔ Migrate the web application to Amazon EC2 instances that run in an Auto Scaling group across two Availability Zones behind an Application Load Balancer. Migrate the database to Amazon RDS with Multi-AZ deployment.

This option is the MOST suitable solution. Using an Auto Scaling group with EC2 instances across multiple Availability Zones ensures high availability and scalability for the web application. Amazon RDS with Multi-AZ deployment provides automatic failover and high availability for the database with minimal administrative overhead.

- ⊗ Migrate the web application to three Amazon EC2 instances across three Availability Zones behind an Application Load Balancer. Migrate the database to three EC2 instances across three Availability Zones.

This option doesn't utilise managed service for the database and requires manual administration of EC2 instances for the database, which creates a high operational overhead. While it can be highly available, it doesn't provide the least administrative overhead compared to the RDS Multi-AZ solution.

Overall Explanation:

The optimal solution involves using managed services to minimize administrative overhead while maximizing availability and scalability. Auto Scaling groups for the web application and RDS with Multi-AZ deployment for the database are key components. Auto Scaling group with EC2 instances behind an ALB provides high availability and scalability. RDS Multi-AZ deployment provides automatic failover, minimizing administrative overhead and ensuring database availability.

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Question 3

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Storage



A company's on-premises business application generates hundreds of files daily, stored on an SMB file share, requiring low-latency connection to application servers. A new policy mandates copying all application-generated files to AWS. A VPN connection to AWS already exists. The application development team lacks time for code modifications to move the application to AWS. Which service should a solutions architect recommend to copy files to AWS?

☐ Amazon Elastic File System (Amazon EFS)

Amazon EFS is a network file system for use with AWS Cloud compute instances and on-premises servers. It does not natively integrate with on-premises SMB shares without modification to the application.

☐ Amazon FSx for Windows File Server

While Amazon FSx for Windows File Server supports SMB, it's primarily designed for cloud-native Windows applications and doesn't directly facilitate copying data from an existing on-premises SMB share without code modifications or additional tools.

☐ AWS Snowball

AWS Snowball is a data transport solution that uses physical devices to move large amounts of data into and out of AWS. It's suitable for initial bulk data migration but isn't designed for continuous, automated file copying.

☒ AWS Storage Gateway

AWS Storage Gateway provides hybrid cloud storage services, offering on-premises access to virtually unlimited cloud storage. The File Gateway configuration supports SMB (and NFS), enabling on-premises applications to seamlessly store and retrieve files in Amazon S3 using existing file system protocols, without application modifications.

Overall Explanation:

AWS Storage Gateway, specifically the File Gateway configuration, is the recommended solution. It allows seamless integration with the existing on-premises SMB file share and automatically copies files to AWS (e.g., Amazon S3) in the background. This meets the requirement of copying all application-generated files to AWS without requiring application code modifications and leverages the existing VPN connection.

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Question 4

Not Attempted

Serverless



A company with 15 employees stores employee start dates in an Amazon DynamoDB table. The company wants to send an email message to each employee on the day of their work anniversary. Which solution will meet these requirements with the MOST operational efficiency?

- ☐ Create a script that scans the DynamoDB table and uses Amazon Simple Notification Service (Amazon SNS) to send email messages to employees when necessary. Use a cron job to run this script every day on an Amazon EC2 instance.
This solution requires managing an EC2 instance and a cron job, increasing operational overhead.
- ☐ Create a script that scans the DynamoDB table and uses Amazon Simple Queue Service (Amazon SQS) to send email messages to employees when necessary. Use a cron job to run this script every day on an Amazon EC2 instance.
This solution requires managing an EC2 instance and a cron job, increasing operational overhead. SQS is also not suitable for sending emails.
- ☒ Create an AWS Lambda function that scans the DynamoDB table and uses Amazon Simple Notification Service (Amazon SNS) to send email messages to employees when necessary. Schedule this Lambda function to run every day.
This solution uses a serverless approach with Lambda and SNS, reducing operational overhead. Lambda can be scheduled to run daily.
- ☐ Create an AWS Lambda function that scans the DynamoDB table and uses Amazon Simple Queue Service (Amazon SQS) to send email messages to employees when necessary. Schedule this Lambda function to run every day.
This solution uses Lambda, but SQS is not designed for sending email messages directly.

Overall Explanation:

The most operationally efficient solution is to use a Lambda function triggered daily to scan the DynamoDB table and send email messages using SNS. This approach avoids the overhead of managing an EC2 instance and cron job. SNS is the correct service to send email. SQS is used for message queuing, not direct email sending.

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Question 5

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[Compute](#)

A company's application is running on Amazon EC2 instances within an Auto Scaling group behind an Elastic Load Balancing (ELB) load balancer. Based on the application's history, the company anticipates a spike in traffic during a holiday each year. A solutions architect must design a strategy to ensure that the Auto Scaling group proactively increases capacity to minimize any performance impact on application users. Which solution will meet these requirements?

- ☐ Create an Amazon CloudWatch alarm to scale up the EC2 instances when CPU utilization exceeds 90%.

This is a reactive approach and does not proactively increase capacity before the expected traffic spike.

- ☒ Create a recurring scheduled action to scale up the Auto Scaling group before the expected period of peak demand.

This is a proactive approach. A scheduled action allows the Auto Scaling group to increase capacity in advance of the anticipated traffic spike.

- ☐ Increase the minimum and maximum number of EC2 instances in the Auto Scaling group during the peak demand period.

While increasing the minimum and maximum sizes has some benefit, it doesn't proactively scale up before the event. It simply allows the Auto Scaling group to scale higher if needed, but won't trigger scaling.

- ☐ Configure an Amazon Simple Notification Service (Amazon SNS) notification to send alerts when there are autoscaling:EC2_INSTANCE_LAUNCH events.

This option only provides notification of scaling events and does not proactively increase capacity.

Overall Explanation:

The best solution is to create a recurring scheduled action. Since the company anticipates a spike in traffic during a holiday each year, a scheduled action is an ideal way to proactively scale up the Auto Scaling group before the expected period of peak demand, minimizing any performance impact. CloudWatch alarms are reactive and trigger scaling based on

metrics, which might be too late. Increasing the min/max capacity only allows for scaling but doesn't trigger it. SNS notifications are for awareness and not proactive scaling.

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Question 6

Not Attempted

Databases



A company uses Amazon RDS for PostgreSQL databases for its data tier. The company is required to implement automated password rotation for these databases with minimal operational overhead. Which solution should the company implement?

- ☒ Store the password in AWS Secrets Manager. Enable automatic rotation on the secret.
AWS Secrets Manager allows for automatic rotation of secrets, including database passwords, with minimal operational overhead. It is specifically designed for this purpose.
- ☐ Store the password in AWS Systems Manager Parameter Store. Enable automatic rotation on the parameter.
AWS Systems Manager Parameter Store does not natively support automatic password rotation. It requires custom solutions like Lambda functions, increasing operational overhead.
- ☐ Store the password in AWS Systems Manager Parameter Store. Write an AWS Lambda function that rotates the password.
While possible, using Parameter Store with a Lambda function for password rotation increases operational overhead due to the need to develop, deploy, and manage the Lambda function.
- ☐ Store the password in AWS Key Management Service (AWS KMS). Enable automatic rotation on the AWS KMS key.
AWS KMS is designed for managing encryption keys, not database passwords. It doesn't offer automatic rotation of database credentials and is not an appropriate choice for this scenario.

Overall Explanation:

AWS Secrets Manager is the best option because it provides built-in functionality for automatic secret rotation, reducing operational overhead. Parameter Store requires a custom implementation with Lambda, and KMS is not designed for managing database passwords. Secrets Manager allows setting up automatic rotation for secrets (passwords).
<https://docs.aws.amazon.com/secretsmanager/latest/userguide/rotating-secrets.html>

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Question 7

Not Attempted

Databases



A company currently runs its application on Oracle Database Enterprise Edition on-premises. The company plans to migrate the application and its database to AWS, leveraging the Bring Your Own License (BYOL) model. A critical requirement is maintaining support for third-party database features that necessitate privileged access. A solutions architect needs to design the most cost-effective solution for this database migration to AWS. Which solution meets these requirements?

- ☐ Migrate the database to Amazon RDS for Oracle by using native tools. Replace the third-party features with AWS Lambda.

This option doesn't directly address the need for privileged access to third-party database features and requires significant application changes. Replacing database features with Lambda functions might not be feasible or cost-effective.

- ☒ Migrate the database to Amazon RDS Custom for Oracle by using native tools. Customize the new database settings to support the third-party features.

Amazon RDS Custom allows for customization of the database environment, enabling support for third-party features that require privileged access. It also supports the BYOL model for Oracle licenses, making it a suitable and cost-effective solution.

- ☐ Migrate the database to Amazon DynamoDB by using AWS Database Migration Service (AWS DMS). Customize the new database settings to support the third-party features.

Amazon DynamoDB is a NoSQL database and is not compatible with Oracle and third-party features, thus this option does not fit the customer's requirements.

- ☐ Migrate the database to Amazon RDS for PostgreSQL by using AWS Database Migration Service (AWS DMS). Rewrite the application code to remove the dependency on third-party features.

This option requires significant application code rewrites, which can be complex and expensive. It also involves migrating to a different database engine (PostgreSQL), which introduces compatibility challenges and is not the most cost-effective solution given the requirements.

Overall Explanation:

Amazon RDS Custom for Oracle allows the necessary level of customization to support third-party features that require privileged access, while also supporting the BYOL model.

This approach avoids the need for costly application code rewrites or the need to migrate to a completely different database platform. Therefore, migrating to Amazon RDS Custom for Oracle is the most cost-effective solution.

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Question 8

Not Attempted

Monitoring & Management



A large organization with a significant presence in the AWS Cloud utilizes Amazon EC2, Amazon RDS, and Amazon DynamoDB. They currently rely on disparate, custom scripts for backing up their infrastructure. To improve efficiency and reduce management overhead, the organization aims to centralize backup management and automate data backups using AWS-native solutions. Which solution BEST meets these requirements?

- ☐ Use third-party backup software with an AWS Storage Gateway tape gateway virtual tape library.

This solution introduces third-party dependencies and doesn't leverage native AWS backup capabilities for EC2, RDS, and DynamoDB directly. While tape gateway can be useful, it doesn't address the need for centralized management and automation with AWS native tools.

- ☒ Use AWS Backup to configure and monitor all backups for the services in use.

AWS Backup is a centralized backup service that supports EC2, RDS, and DynamoDB, providing a single pane of glass for managing and automating backups. It allows for policy-based backup scheduling and retention, meeting the requirements of centralized management and automation.

- ☐ Use AWS Config to set lifecycle management to take snapshots of all data sources on a schedule.

AWS Config is primarily a configuration management service and not a backup service. While it can trigger actions based on configuration changes, it does not provide centralized backup management or automated scheduling of backups across all specified services.

- ☐ Use AWS Systems Manager State Manager to manage the configuration and monitoring of backup tasks.

AWS Systems Manager State Manager is a configuration management service, not a dedicated backup service. While it can be used to automate tasks like running backup scripts, it doesn't provide the centralized management and automation capabilities offered by AWS Backup for various data sources.

Overall Explanation:

AWS Backup is the ideal solution because it's an AWS-native service designed for centralized management and automation of backups for various AWS services, including EC2, RDS, and DynamoDB. It provides a single console to configure, manage, and monitor backups, fulfilling the organization's requirements to move away from custom scripts and streamline their backup processes. The other options either introduce third-party dependencies or are not specifically designed for centralized backup management.

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Question 9

Not Attempted

Storage



A company has a production web application where users upload documents through a web interface or a mobile app. Due to a new regulatory requirement, new documents must not be modified or deleted after they are stored. Which solution should a solutions architect implement to meet this requirement?

- ☒ Store the uploaded documents in an Amazon S3 bucket with S3 Versioning and S3 Object Lock enabled.
S3 Versioning preserves every version of a document, preventing accidental or intentional modifications/deletions. S3 Object Lock allows setting a retention period or legal hold on objects, making them immutable, satisfying the regulatory requirement.
- ☐ Store the uploaded documents in an Amazon S3 bucket. Configure an S3 Lifecycle policy to archive the documents periodically.
Configuring an S3 Lifecycle policy to archive documents periodically does not guarantee the prevention of document modification or deletion after they are stored. Archiving is for cost optimization, not immutability.
- ☐ Store the uploaded documents in an Amazon S3 bucket with S3 Versioning enabled. Configure an ACL to restrict all access to read-only.
Enabling S3 Versioning alone does not prevent modifications or deletions of objects. Configuring an ACL does not guarantee the prevention of modifications or deletions by authorized users with sufficient permissions.
- ☐ Store the uploaded documents on an Amazon Elastic File System (Amazon EFS) volume. Access the data by mounting the volume in read-only mode.

Using EFS does not prevent modifications or deletions of the documents by users or processes with write permissions. Mounting in read-only mode relies on the application to enforce restrictions, which is not as robust as using S3 Object Lock.

Overall Explanation:

The best solution is to use Amazon S3 with S3 Versioning and S3 Object Lock. S3 Versioning ensures that all versions of an object are retained, while S3 Object Lock prevents objects from being overwritten or deleted for a specified period or indefinitely using a WORM (Write Once Read Many) model. This combination satisfies the regulatory requirement of preventing modifications and deletions.

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Question 10

Not Attempted

Security



A company aims to establish a comprehensive map of its IT infrastructure to effectively identify and enforce policies on resources that present security vulnerabilities. The security team requires the ability to query this IT infrastructure map and promptly detect security risks with minimal administrative effort. Which solution best fulfills these requirements while minimizing operational overhead?

☐ Use Amazon RDS to store the data. Use SQL to query the data to identify security risks.
Amazon RDS is a relational database service, which is not optimized for graph data representing relationships between resources in an IT infrastructure. SQL queries would be complex and inefficient for traversing these relationships.

☒ Use Amazon Neptune to store the data. Use SPARQL to query the data to identify security risks.

Amazon Neptune is a graph database service ideal for storing and querying interconnected data, such as an IT infrastructure map. SPARQL is a query language designed for graph databases, enabling efficient identification of security risks by traversing relationships.

☐ Use Amazon Redshift to store the data. Use SQL to query the data to identify security risks.

Amazon Redshift is a data warehouse service optimized for analytical workloads. It is not designed for real-time querying of graph data representing an IT infrastructure map. SQL queries would be complex and inefficient.

- ⊗ Use Amazon DynamoDB to store the data. Use PartiQL to query the data to identify security risks.

Amazon DynamoDB is a NoSQL database service. While PartiQL allows SQL-compatible queries, it is not optimized for traversing complex relationships in graph data. Managing relationships would be more complex compared to a graph database.

Overall Explanation:

Amazon Neptune is the best solution because it's a graph database specifically designed for storing and querying relationship-heavy data like IT infrastructure maps. SPARQL is the standard query language for graph databases, making it efficient for identifying security risks based on resource relationships. This combination results in the least operational overhead for the given requirements.

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Question 11

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Databases



A large company requires a cost-effective solution to provide its globally located developers with separate, limited-size, managed PostgreSQL databases for development purposes. These databases will have low volume and developers need them only when actively working. Which solution meets these requirements MOST cost-effectively?

- ⊗ Give the developers the ability to launch separate Amazon Aurora instances. Set up a process to shut down Aurora instances at the end of the workday and to start Aurora instances at the beginning of the next workday.
- This option is more expensive than a serverless solution because Amazon Aurora instances use On-Demand or Reserved instances.
- ⊗ Develop an AWS Service Catalog product that enforces size restrictions for launching Amazon Aurora instances. Give the developers access to launch the product when they need a development database.
- This option is more expensive than a serverless solution because Amazon Aurora instances use On-Demand or Reserved instances.
- ✔ Create an Amazon Aurora Serverless cluster. Develop an AWS Service Catalog product to launch databases in the cluster with the default capacity settings. Grant the developers access to the product.

Amazon Aurora Serverless automatically starts up, shuts down, and scales capacity based on application needs. You only pay for consumed capacity, making it cost-effective. AWS Service Catalog provides a catalog of services that users can utilize within permitted configurations.

- ⊗ Monitor AWS Trusted Advisor checks for idle Amazon RDS databases. Create a process to terminate identified idle RDS databases.

This option involves manual monitoring and termination, introducing operational overhead and potentially causing delays in database availability for developers. Furthermore, RDS instances are more expensive than Aurora Serverless.

Overall Explanation:

Amazon Aurora Serverless combined with AWS Service Catalog is the most cost-effective solution. Aurora Serverless automatically scales and shuts down, reducing costs. AWS Service Catalog allows developers to easily access the required resources within predefined configurations. Using Aurora or RDS instances involve on-demand or reserved instances. Hence, option A, B and D are less cost-effective. Option C provides an automated and cost optimized solution.

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Question 12

Not Attempted

Compute



A company hosts a content management system (CMS) on Amazon EC2 instances behind an Application Load Balancer (ALB). These instances reside within an Auto Scaling group spanning multiple Availability Zones. Users frequently update website content, requiring all EC2 instances to maintain synchronized assets with minimal delay. What solution ensures efficient and real-time content sharing across all EC2 instances with the least possible lag time?

- ⊗ Update the EC2 user data in the Auto Scaling group lifecycle policy to copy website assets from the newest EC2 instance. Configure the ALB to direct changes only to this instance.

This solution has significant lag time and potential inconsistencies as the content is only synchronized during new instance launches. It does not address ongoing updates.

- ✓ Copy the website assets to an Amazon Elastic File System (Amazon EFS) file system. Configure each EC2 instance to mount the EFS file system locally and reference the assets within.

Amazon EFS provides a shared file system accessible by all EC2 instances. This allows for real-time content updates to be immediately available across all instances, fulfilling the requirement.

- ⊗ Copy website assets to an Amazon S3 bucket. Configure each EC2 instance to download assets from S3 to an attached Amazon EBS volume. Use a scheduled S3 sync command to maintain updates.

Using S3 and a sync command introduces significant delay and complexity, as content is only updated hourly, failing to meet the near real-time requirement. EBS volumes are also not inherently shared between instances.

- ⊗ Restore an Amazon Elastic Block Store (Amazon EBS) snapshot with website assets. Attach the EBS snapshot as a secondary EBS volume to new EC2 instances. Configure the website application to reference the assets on this volume.

This approach only provides the initial website data and doesn't account for updates. EBS snapshots aren't designed for real-time content sharing, and each instance would have a static copy of the data.

Overall Explanation:

Amazon EFS (Elastic File System) is the ideal solution for sharing website assets across multiple EC2 instances in real-time. By mounting the EFS file system on each instance, content updates are immediately reflected across all instances, ensuring minimal lag time and data consistency. Using user data to copy content or using S3 sync methods introduces significant delays and is not suitable for a dynamic content management system requiring real-time updates. Therefore, EFS ensures the content is up-to-date and readily available for all EC2 instances without significant latency.

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Question 13

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Security



A company's web application consists of multiple Amazon EC2 instances that run behind an Application Load Balancer in a VPC. An Amazon RDS for MySQL DB instance contains the data. The company needs the ability to automatically detect and respond to suspicious or unexpected behavior in its AWS environment. The company already has added AWS WAF to its architecture. What should a solutions architect do next to protect against threats?

- ✔ Use Amazon GuardDuty to perform threat detection. Configure Amazon EventBridge to filter for GuardDuty findings and to invoke an AWS Lambda function to adjust the AWS WAF rules.

Amazon GuardDuty is a threat detection service that continuously monitors for malicious activity and unauthorized behavior across your AWS environment. EventBridge can be used to trigger Lambda functions based on GuardDuty findings to update WAF rules automatically.

- ✘ Use AWS Firewall Manager to perform threat detection. Configure Amazon EventBridge to filter for Firewall Manager findings and to invoke an AWS Lambda function to adjust the AWS WAF web ACL.

AWS Firewall Manager is primarily used for managing AWS WAF rules across multiple accounts and resources. While it provides centralized management, GuardDuty is a more appropriate service for threat detection.

- ✘ Use Amazon Inspector to perform threat detection and to update the AWS WAF rules. Create a VPC network ACL to limit access to the web application.

Amazon Inspector is a vulnerability management service that assesses EC2 instances and container images for software vulnerabilities and unintended network exposure. It does not directly detect threats like GuardDuty does, nor does it directly integrate with WAF for dynamic rule updates. While network ACLs enhance security, they are not the primary solution for threat detection.

- ✘ Use Amazon Macie to perform threat detection and to update the AWS WAF rules. Create a VPC network ACL to limit access to the web application.

Amazon Macie is a data security and privacy service that uses machine learning and pattern matching to discover and protect sensitive data in AWS. It is not primarily designed for general threat detection or integration with WAF.

Overall Explanation:

Amazon GuardDuty is designed for threat detection, continuously monitoring for malicious activity and unauthorized behavior. Configuring Amazon EventBridge to react to GuardDuty findings and trigger a Lambda function to adjust AWS WAF rules provides an automated response to identified threats.

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Question 14

Not Attempted

Databases



A company is planning to run a group of Amazon EC2 instances that connect to an Amazon Aurora database. The company has built an AWS CloudFormation template to deploy the EC2 instances and the Aurora DB cluster. The company wants to allow the instances to authenticate to the database in a secure way, and does not want to maintain static database credentials. Which solution meets these requirements with the LEAST operational effort?

- ☐ Create a database user with a user name and password. Add parameters for the database user name and password to the CloudFormation template. Pass the parameters to the EC2 instances when the instances are launched.
This solution requires managing credentials and is not the least operational effort.
- ☐ Create a database user with a user name and password. Store the user name and password in AWS Systems Manager Parameter Store. Configure the EC2 instances to retrieve the database credentials from Parameter Store.
This solution still requires managing credentials, although stored securely, and is not the least operational effort.
- ☒ Configure the DB cluster to use IAM database authentication. Create a database user to use with IAM authentication. Associate a role with the EC2 instances to allow applications on the instances to access the database.
This is the correct answer. Using IAM database authentication and associating a role with the EC2 instances is the least operational effort. It avoids managing database passwords.
- ☐ Configure the DB cluster to use IAM database authentication with an IAM user. Create a database user that has a name that matches the IAM user. Associate the IAM user with the EC2 instances to allow applications on the instances to access the database.
You cannot directly associate an IAM user with EC2 instances. You associate an IAM role, making this option incorrect.

Overall Explanation:

The best solution is to configure IAM database authentication for the Aurora cluster and associate an IAM role with the EC2 instances. This allows the instances to authenticate to the database without needing to store or manage database credentials. This is the least operational effort because it leverages IAM roles for authentication, eliminating the need for static credentials. IAM roles can be easily managed and rotated by AWS and don't require manual rotation or management.

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Question 15

Not Attempted

Networking



A company wants to configure its Amazon CloudFront distribution to use SSL/TLS certificates with a custom domain name. The company wants to deploy the certificate without incurring additional costs. Which solution meets these requirements?

- ☐ Request an Amazon issued private certificate from AWS Certificate Manager (ACM) in the us-east-1 Region.
Private certificates are generally used for internal resources and are not suitable for CloudFront distributions with custom domain names. Also, Private CA is not free.
- ☐ Request an Amazon issued private certificate from AWS Certificate Manager (ACM) in the us-west-1 Region.
Private certificates are generally used for internal resources and are not suitable for CloudFront distributions with custom domain names. ACM certificates for CloudFront must be requested in the us-east-1 Region. Also, Private CA is not free.
- ☒ Request an Amazon issued public certificate from AWS Certificate Manager (ACM) in the us-east-1 Region.
ACM provides free public SSL/TLS certificates. To use an ACM certificate with CloudFront, it must be requested in the US East (N. Virginia) Region (us-east-1).
- ☐ Request an Amazon issued public certificate from AWS Certificate Manager (ACM) in the us-west-1 Region.
While public certificates from ACM are free, ACM certificates for CloudFront must be requested in the us-east-1 Region.

Overall Explanation:

Public SSL/TLS certificates provisioned through AWS Certificate Manager (ACM) are free. To use an ACM certificate with CloudFront, you must request it in the US East (N. Virginia) Region (us-east-1). Therefore, requesting an Amazon issued public certificate from ACM in us-east-1 is the correct solution.

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Question 16

Not Attempted

Security



A company stores operations data in an Amazon S3 bucket. For an annual audit, an external consultant requires access to an annual report stored in the bucket for 7 days. The company needs a solution that grants the consultant access only to the specific report with the highest operational efficiency. Which solution meets these requirements?

- ☐ Create a new S3 bucket configured to host a public static website. Migrate the report to the new bucket and share the website URL with the consultant.
Creating a new bucket and migrating the report is operationally inefficient and unnecessarily exposes the report via a public website.
- ☐ Enable public access to the entire S3 bucket for 7 days. Remove public access when the consultant completes the audit.
Enabling public access to the entire bucket is a security risk and grants the consultant more access than necessary.
- ☐ Create a new IAM user with access to the report in the S3 bucket. Provide the access keys to the consultant. Revoke the access keys after 7 days.
Creating and managing an IAM user adds unnecessary operational overhead compared to using a presigned URL.
- ☒ Generate a presigned URL that provides the required access to the specific report in the S3 bucket. Share the presigned URL with the consultant.
Presigned URLs provide secure, temporary access to specific S3 objects without requiring IAM user creation or broad public access.

Overall Explanation:

A presigned URL provides temporary, secure access to a specific object in S3. It is the most operationally efficient solution because it avoids creating new IAM users or modifying bucket-level permissions. The consultant gains access only to the required report for the specified duration (7 days) without any ongoing management overhead. Presigned URLs are ideal for providing short-term access to private objects in S3 by appending an AWS Access Key, expiration time, and Sigv4 signature as query parameters to the S3 object.

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Question 17

Not Attempted

Networking



A company is planning to run a High Performance Computing (HPC) workload on Amazon EC2 instances. The HPC workload demands low-latency network performance and high network throughput for tightly coupled node-to-node communication. Which of the following solutions will best satisfy these requirements?

- ☒ **Configure the EC2 instances to be part of a cluster placement group.**
Cluster placement groups provide low latency and high network throughput which are essential for tightly coupled node-to-node communication in HPC workloads.
- ☐ **Launch the EC2 instances with Dedicated Instance tenancy.**
Dedicated Instance tenancy ensures that the instances run on dedicated hardware, but it does not directly address the need for low-latency or high-throughput networking.
- ☐ **Launch the EC2 instances as Spot Instances.**
Spot Instances offer cost savings, but they are subject to interruption and do not guarantee low latency or high network throughput.
- ☐ **Configure an On-Demand Capacity Reservation when the EC2 instances are launched.**
On-Demand Capacity Reservations guarantee the availability of EC2 capacity but do not inherently provide low-latency or high-throughput networking.

Overall Explanation:

Cluster placement groups are designed to provide low-latency and high-bandwidth connectivity between instances within the group, which is essential for HPC workloads that require tightly coupled node-to-node communication. Dedicated Instances offer dedicated hardware, but do not focus on low latency and high throughput. Spot Instances are about pricing. On-Demand Capacity Reservations are for ensuring capacity is available, but do not ensure high networking performance.

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Question 18

Not Attempted

Networking



A company has primary and secondary data centers that are 500 miles (804.7 km) apart and interconnected with high-speed fiber-optic cable. The company requires a highly available and secure network connection between its data centers and a VPC on AWS for a mission-critical workload. To ensure maximum resiliency, which connectivity solution should a solutions architect recommend?

- ☐ Two AWS Direct Connect connections from the primary data center terminating at two Direct Connect locations on two separate devices.

This option only provides redundancy for the primary data center, not both. It doesn't protect against failure in the secondary data center.

- ☐ A single AWS Direct Connect connection from each of the primary and secondary data centers terminating at one Direct Connect location on the same device.

This option introduces a single point of failure at the Direct Connect location and device.

- ☒ Two AWS Direct Connect connections from each of the primary and secondary data centers terminating at two Direct Connect locations on two separate devices.

This option provides maximum redundancy by having multiple connections from both data centers terminating at different Direct Connect locations on separate devices. This protects against data center failure, connection failure, and Direct Connect location/device failure.

- ☐ A single AWS Direct Connect connection from each of the primary and secondary data centers terminating at one Direct Connect location on two separate devices.

While this provides some redundancy with two separate devices, it introduces a single point of failure at the Direct Connect location. If the Direct Connect location experiences issues, connectivity from both data centers will be impacted.

Overall Explanation:

The most resilient solution is to have two independent Direct Connect connections from each data center, terminating at two different Direct Connect locations on separate devices. This setup provides redundancy against data center failures, connection failures, and Direct Connect location/device failures. Option C ensures maximum availability for the mission-critical workload.

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A company runs several Amazon RDS for Oracle On-Demand DB instances that have high utilization. The RDS DB instances run in member accounts that are in an organization in AWS Organizations. The company's finance team has access to the organization's management account and member accounts. The finance team wants to find ways to optimize costs by using AWS Trusted Advisor. Which combination of steps will meet these requirements? (Choose two.)

- ☐ Use the Trusted Advisor recommendations in the management account.
Trusted Advisor in the management account doesn't provide insights for member accounts unless the organizational view is enabled.
- ☒ Use the Trusted Advisor recommendations in the member accounts where the RDS DB instances are running.
The finance team needs access to each member account to review Trusted Advisor recommendations directly, which might be operationally complex.
- ☒ Review the Trusted Advisor checks for Amazon RDS Reserved Instance Optimization.
Given the high utilization of On-Demand instances, reviewing the Reserved Instance Optimization checks is crucial to identify potential cost savings by switching to Reserved Instances.
- ☐ Review the Trusted Advisor checks for Amazon RDS Idle DB Instances.
Because the RDS instances have high utilization, this check is less relevant.
- ☐ Review the Trusted Advisor checks for compute optimization. Crosscheck the results by using AWS Compute Optimizer.
The question explicitly mentions RDS DB instances and optimizing their costs. Compute Optimizer primarily focuses on EC2 instances and not RDS.

Overall Explanation:

The best approach is to review the Trusted Advisor recommendations in the member accounts directly to get specific insights for each RDS instance. Additionally, because the instances have high utilization and are On-Demand, checking the Reserved Instance Optimization is a must to save money. Options A, D and E are less relevant or incorrect. A is not applicable as Trusted Advisor is account specific unless organizational view is enabled, D is wrong as the instances are highly utilized, E is irrelevant as it is specific to compute optimization and not RDS.

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Question 20

Not Attempted

Security



A company has several web servers that need to frequently access a common Amazon RDS MySQL Multi-AZ DB instance. The company wants a secure method for the web servers to connect to the database while meeting a security requirement to rotate user credentials frequently. Which solution meets these requirements?

- ☒ Store the database user credentials in AWS Secrets Manager. Grant the necessary IAM permissions to allow the web servers to access AWS Secrets Manager.
Secrets Manager allows secure storage and automatic rotation of credentials, and IAM permissions grant access to the web servers.
- ☐ Store the database user credentials in AWS Systems Manager OpsCenter. Grant the necessary IAM permissions to allow the web servers to access OpsCenter.
SSM OpsCenter is not designed for storing and rotating database credentials; it's primarily for operational issue management.
- ☐ Store the database user credentials in a secure Amazon S3 bucket. Grant the necessary IAM permissions to allow the web servers to retrieve credentials and access the database.
Storing credentials in S3 lacks the features and security controls of AWS Secrets Manager for secure storage and rotation.
- ☐ Store the database user credentials in files encrypted with AWS Key Management Service (AWS KMS) on the web server file system. The web server should be able to decrypt the files and access the database.
Managing credentials directly on web servers, even with KMS, introduces complexities and security risks for rotation and scaling.

Overall Explanation:

AWS Secrets Manager is designed for storing and managing secrets such as database credentials. It provides automated secret rotation and IAM integration for secure access, meeting the requirements for secure access and frequent credential rotation. Storing credentials on the web servers or in S3 is less secure and harder to manage. OpsCenter is for operational issue management, not credential storage.

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Question 21

Not Attempted

Networking



A solutions architect is designing an application that will run on Amazon EC2 instances within private subnets across multiple Availability Zones in a VPC. These EC2 instances need to frequently access large files containing confidential data stored in Amazon S3 buckets for processing. To minimize data transfer costs and ensure secure access to S3, what network architecture should the solutions architect implement?

- ☒ Create a gateway endpoint for Amazon S3 in the VPC. In the route tables for the private subnets, add an entry for the gateway endpoint.

A gateway endpoint for S3 provides cost-effective and secure access to S3 from private subnets without traversing the internet. It keeps traffic within the AWS network, thus minimizing data transfer costs and improving security.

- ☐ Create a single NAT gateway in a public subnet. In the route tables for the private subnets, add a default route that points to the NAT gateway.

Using a NAT gateway allows the EC2 instances to access the internet, including S3, but it incurs data transfer costs and requires the traffic to traverse the internet, which is less secure.

- ☐ Create an AWS PrivateLink interface endpoint for Amazon S3 in the VPC. In the route tables for the private subnets, add an entry for the interface endpoint.

While AWS PrivateLink provides secure, private connectivity to AWS services, it is generally more expensive than using a gateway endpoint for S3 access and is more suitable when you need to access the S3 API rather than direct object access.

- ☐ Create one NAT gateway for each Availability Zone in public subnets. In each of the route tables for the private subnets, add a default route that points to the NAT gateway in the same Availability Zone.

Using multiple NAT gateways improves availability but does not address the requirement of minimizing data transfer costs and is also more expensive than using gateway endpoints. Also, the traffic will traverse internet to reach S3.

Overall Explanation:

The most cost-effective and secure solution is to create a gateway endpoint for Amazon S3 in the VPC. This allows the EC2 instances in private subnets to access S3 directly without incurring data transfer costs associated with NAT gateways or exposing the traffic to the

internet. Adding an entry for the gateway endpoint in the route tables ensures that traffic destined for S3 is routed through the endpoint.

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Question 22

Not Attempted

Databases



A company is migrating its on-premises MySQL database to AWS. The database frequently receives data imports from a client application, resulting in a high volume of write operations. The company is concerned about potential performance impacts on the application due to this traffic. How should a solutions architect design the architecture on AWS to address these concerns?

- ☒ Provision an Amazon RDS for MySQL DB instance with Provisioned IOPS SSD storage. Monitor write operation metrics by using Amazon CloudWatch. Adjust the provisioned IOPS if necessary.

Provisioned IOPS storage is designed for I/O-intensive workloads like this. CloudWatch monitoring allows for proactive adjustments to maintain performance.

- ☐ Provision an Amazon RDS for MySQL DB instance with General Purpose SSD storage. Place an Amazon ElastiCache cluster in front of the DB instance. Configure the application to query ElastiCache instead.

ElastiCache is primarily for read caching. While it can reduce read load, it doesn't directly address high write volume and introduces complexity.

- ☐ Provision an Amazon DocumentDB (with MongoDB compatibility) instance with a memory optimized instance type. Monitor Amazon CloudWatch for performance-related issues. Change the instance class if necessary.

Changing the database engine to DocumentDB is not recommended if the current database is MySQL and is working fine. This option is also not optimized to support High IOPS.

- ☐ Provision an Amazon Elastic File System (Amazon EFS) file system in General Purpose performance mode. Monitor Amazon CloudWatch for IOPS bottlenecks. Change to Provisioned Throughput performance mode if necessary.

EFS is a file storage service and not suitable as a database.

Overall Explanation:

Given the high volume of write operations, the most effective solution is to use Amazon RDS for MySQL with Provisioned IOPS SSD storage. This option is optimized for I/O-intensive workloads. Monitoring with CloudWatch ensures that the provisioned IOPS can be adjusted as needed to maintain optimal performance. Using ElastiCache for write-heavy operations is not the most efficient approach. Switching to Amazon DocumentDB without a clear requirement is not recommended, and EFS is not an appropriate storage solution for a relational database.

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Question 23

Not Attempted

Security



A company runs an application in the AWS Cloud that generates sensitive archival data files. The company wants to rearchitect the application's data storage. The company wants to encrypt the data files and to ensure that third parties do not have access to the data *before* the data is encrypted and sent to AWS. The company has already created an Amazon S3 bucket. Which solution will meet these requirements?

- ☐ ☒ Configure the S3 bucket to use client-side encryption with an Amazon S3 managed encryption key. Configure the application to use the S3 bucket to store the archival files.
This option uses client-side encryption, but with an S3-managed key, which means the encryption happens on the server side, not before the data reaches AWS. Thus, it does not satisfy the requirement that third parties should not be able to access the unencrypted data before it reaches AWS.
- ☐ ☒ Configure the S3 bucket to use server-side encryption with AWS KMS keys (SSE-KMS). Configure the application to use the S3 bucket to store the archival files.
Server-side encryption encrypts the data *after* it is uploaded to S3. The requirement is to encrypt the data *before* sending it to AWS.
- ☐ ☒ Configure the S3 bucket to use dual-layer server-side encryption with AWS KMS keys (SSE-KMS). Configure the application to use the S3 bucket to store the archival files.
Dual-layer server-side encryption, like regular server-side encryption, encrypts the data *after* it is uploaded to S3. The requirement is to encrypt the data *before* sending it to AWS.

- ✓ Configure the application to use client-side encryption with a key stored in AWS Key Management Service (AWS KMS). Configure the application to store the archival files in the S3 bucket.

This option uses client-side encryption. The application encrypts the data before it is sent to S3. The encryption key is stored and managed by AWS KMS. This ensures that third parties do not have access to the data before it reaches AWS.

Overall Explanation:

The requirement is to ensure that the data is encrypted *before* it is sent to AWS to prevent third-party access. Only client-side encryption satisfies this requirement. Server-side encryption encrypts the data after it reaches AWS. Using client-side encryption with a key stored in AWS KMS fulfills the requirement.

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Question 24

Not Attempted

Databases



A company uses Amazon RDS with default backup settings for its database tier. The company needs to make a daily backup of the database to meet regulatory requirements. The company must retain the backups for 30 days. Which solution will meet these requirements with the LEAST operational overhead?

- ✗ Write an AWS Lambda function to create an RDS snapshot every day.
This option requires writing and maintaining a Lambda function, increasing operational overhead.
- ✓ Modify the RDS database to have a retention period of 30 days for automated backups.
Amazon RDS performs a full daily backup of your data during a backup window that you define when you create the DB instance. You can configure a retention period of up to 35 days for the automated backup, so simply modifying the retention period is the easiest and least overhead approach.
- ✗ Use AWS Systems Manager Maintenance Windows to modify the RDS backup retention period.
While Systems Manager Maintenance Windows can automate tasks, it's an unnecessarily complex solution for simply changing the RDS backup retention period.

- ⊗ Create a manual snapshot every day by using the AWS CLI. Modify the RDS backup retention period.

Creating daily manual snapshots using the AWS CLI adds significant operational overhead. Additionally, it's not necessary to *also* modify the RDS backup retention period, which is confusing, and already handled automatically by the RDS service.

Overall Explanation:

Amazon RDS automatically performs a full daily snapshot of your data during a backup window that you define when you create the DB instance. You can configure a retention period of up to 35 days for the automated backup. Therefore, modifying the RDS database to have a retention period of 30 days for automated backups provides the required functionality with the least operational overhead.

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Question 25

Not Attempted

Databases



A company runs its application on AWS using an Amazon Aurora DB cluster. During peak hours, numerous users access and read data, leading to degraded write query performance. The company aims to improve application scalability to handle these peak demands cost-effectively. Which solution BEST addresses these requirements?

- ⊗ Create a second Aurora DB cluster. Configure a copy job to replicate the user data to the new database. Update the application to use the second database for read operations.

Creating a second full Aurora cluster and replicating data is a costly solution. Read replicas are more cost-effective for read scaling.

- ⊗ Create an Amazon DynamoDB Accelerator (DAX) cluster in front of the existing Aurora DB cluster. Update the application to use the DAX cluster for read-only queries, writing data directly to the Aurora DB cluster.

DAX is designed for DynamoDB, not Aurora. It cannot be used to cache reads for Aurora.

- ✔ Create an Aurora read replica within the existing Aurora DB cluster. Update the application to use the replica endpoint for read-only queries and the cluster endpoint for write queries.

This is the most cost-effective solution. Read replicas offload read operations from the primary instance, improving write performance. They are also purpose-built for Aurora, ensuring compatibility and optimal performance.

- ⊗ Create an Amazon Redshift cluster. Copy the user data to the Redshift cluster. Update the application to connect to the Redshift cluster and perform read-only queries.

Redshift is a data warehouse solution and is not ideal for transactional read operations of an application. It's better suited for analytics and reporting.

Overall Explanation:

Creating an Aurora read replica (Option C) is the MOST cost-effective solution. It allows you to offload read queries from the primary database instance, thus improving write performance during peak usage. Using read replicas is the recommended approach for scaling read capacity in Aurora clusters without incurring the high costs of creating separate full clusters (Option A) or misusing services like DAX (designed for DynamoDB (Option B)) or Redshift (designed for data warehousing (Option D)).

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Question 26

Not Attempted

Serverless



A company's near-real-time streaming application is running on AWS. As the data is ingested, a job runs on the data and takes 30 minutes to complete. The workload frequently experiences high latency due to large amounts of incoming data. A solutions architect needs to design a scalable and serverless solution to enhance performance. Which combination of steps should the solutions architect take? (Choose two.)

- ✔ Use Amazon Kinesis Data Firehose to ingest the data.
Kinesis Data Firehose is designed for ingesting streaming data into AWS data stores and analytics services.
- ⊗ Use AWS Lambda with AWS Step Functions to process the data.
Lambda functions have a maximum execution time of 15 minutes, which is insufficient for the 30-minute job. Step Functions manages workflows but doesn't solve the Lambda execution time limitation.
- ⊗ Use AWS Database Migration Service (AWS DMS) to ingest the data.

AWS DMS is for database migrations, not general streaming data ingestion.

- ☐ Use Amazon EC2 instances in an Auto Scaling group to process the data.
While EC2 with Auto Scaling can provide scalability, it is not a serverless solution.
- ☒ Use AWS Fargate with Amazon Elastic Container Service (Amazon ECS) to process the data.
AWS Fargate with ECS allows you to run containers without managing servers, offering a scalable and serverless solution capable of running longer-duration jobs.

Overall Explanation:

To build a scalable and serverless solution, Kinesis Data Firehose is used for ingesting the streaming data (Option A). Since the job takes 30 minutes, Lambda is not suitable as it has a 15-minute execution time limit. ECS with Fargate provides a serverless compute environment to run the 30-minute job (Option E).

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Question 27

Not Attempted

Security



A company runs a web application on multiple Amazon EC2 instances in a VPC. The application needs to write sensitive data to an Amazon S3 bucket. The data cannot be sent over the public internet. Which solution will meet these requirements?

- ☒ Create a gateway VPC endpoint for Amazon S3. Create a route in the VPC route table to the endpoint.
A gateway VPC endpoint for S3 allows traffic to stay within the AWS network and does not traverse the public internet. The route table entry directs traffic destined for S3 to the endpoint.
- ☐ Create an internal Network Load Balancer that has the S3 bucket as the target.
An internal Network Load Balancer cannot target an S3 bucket directly. S3 is not a compute instance or IP address that can be targeted by an NLB.
- ☐ Deploy the S3 bucket inside the VPC. Create a route in the VPC route table to the bucket.
It's not possible to deploy an S3 bucket directly inside a VPC. S3 is a service that resides outside of your VPC, but can be accessed securely from within it.

- ⊗ Create an AWS Direct Connect connection between the VPC and an S3 regional endpoint.

While Direct Connect provides a dedicated network connection to AWS, it doesn't inherently prevent data from traversing the public internet unless combined with private VIF and appropriate routing. A VPC endpoint is a more direct and cost-effective solution for keeping S3 traffic within the AWS network.

Overall Explanation:

A gateway VPC endpoint for Amazon S3 is the correct solution because it allows the EC2 instances in the VPC to access S3 without traversing the public internet. The endpoint creates a route within the VPC's network to S3. The other options are incorrect because they either don't prevent internet traffic, can't be implemented, or are more complex than necessary. Direct connect is a solution for hybrid environment.

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Question 28

Not Attempted

Cost Optimization



A company runs its production workload on Amazon EC2 instances with Amazon Elastic Block Store (Amazon EBS) volumes. A solutions architect needs to analyze the current EBS volume cost and to recommend optimizations. The recommendations need to include estimated monthly saving opportunities. Which solution will meet these requirements?

- ⊗ Use Amazon Inspector reporting to generate EBS volume recommendations for optimization.
- Amazon Inspector is a vulnerability management service and does not provide EBS volume cost analysis or optimization recommendations.
- ⊗ Use AWS Systems Manager reporting to determine EBS volume recommendations for optimization.
- AWS Systems Manager provides operational insights and automation capabilities, but it's not the primary service for EBS cost optimization recommendations.
- ⊗ Use Amazon CloudWatch metrics reporting to determine EBS volume recommendations for optimization.
- Amazon CloudWatch provides metrics and monitoring, but requires manual analysis and doesn't directly generate optimization recommendations with cost savings estimates.

- ✓ Use AWS Compute Optimizer to generate EBS volume recommendations for optimization.

AWS Compute Optimizer analyzes the utilization metrics of your AWS resources, including EBS volumes, and provides recommendations to optimize performance and reduce costs, including estimated monthly savings.

Overall Explanation:

AWS Compute Optimizer provides recommendations to optimize your use of AWS resources, including EC2 instances and EBS volumes. It helps avoid over-provisioning and under-provisioning based on your utilization data and can estimate monthly saving opportunities. Therefore, it is the most appropriate solution for analyzing EBS volume costs and recommending optimizations.

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Question 29

Not Attempted

Serverless



A company needs to enhance its e-commerce order-processing application, currently deployed on AWS. The application must guarantee that each order is processed exactly once and maintain a positive customer experience, even during unpredictable traffic surges. Which solution BEST meets these requirements?

- ✓ Create an Amazon Simple Queue Service (Amazon SQS) FIFO queue. Place all orders into the SQS queue. Configure an AWS Lambda function as the target to process the orders.
- Using an SQS FIFO queue guarantees that messages (orders) are processed exactly once, and in the order they were received. Lambda can scale to handle unpredictable traffic surges.
- ✗ Create an Amazon Simple Notification Service (Amazon SNS) standard topic. Publish all orders to the SNS standard topic. Configure the application as a notification target.
- SNS standard topics do not guarantee message order or exactly-once processing. Messages could be lost or delivered more than once, which violates the requirements.
- ✗ Create a flow by using Amazon AppFlow. Send the orders to the flow. Configure an AWS Lambda function as the target to process the orders.
- Amazon AppFlow is designed to transfer data between AWS services and SaaS applications. It's not primarily designed for guaranteeing exactly-once processing of e-commerce orders in a queueing

scenario.

- ⊗ Configure AWS X-Ray in the application to track the order requests. Configure the application to process the orders by pulling the orders from Amazon CloudWatch. AWS X-Ray is for tracing and debugging distributed applications. CloudWatch is primarily for monitoring and logging. Neither service provides the queueing and exactly-once processing required.

Overall Explanation:

The best solution is to use an Amazon SQS FIFO queue. FIFO queues guarantee that each message (order) is processed exactly once, and in the same order it was sent. Utilizing an AWS Lambda function as a consumer for the SQS queue provides scalability to handle unpredictable traffic surges without affecting the customer experience.

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Question 30

Not Attempted

Databases



A company hosts an application on AWS Lambda functions that are invoked by an Amazon API Gateway API. The Lambda functions save customer data to an Amazon Aurora MySQL database. Whenever the company upgrades the database, the Lambda functions fail to establish database connections until the upgrade is complete, resulting in customer data not being recorded for some events. A solutions architect needs to design a solution that stores customer data created during database upgrades. Which solution will meet these requirements?

- ⊗ Provision an Amazon RDS Proxy to sit between the Lambda functions and the database. Configure the Lambda functions to connect to the RDS Proxy. RDS Proxy primarily helps manage database connections but does not inherently store data during outages or upgrades. While it can reduce failover times, it doesn't guarantee data persistence during prolonged database unavailability.
- ⊗ Increase the run time of the Lambda functions to the maximum. Create a retry mechanism in the code that stores the customer data in the database. Increasing Lambda runtime and adding retry logic might help with transient errors, but it's insufficient if the database upgrade duration exceeds Lambda's maximum execution time. Data could still be lost.

- ⊗ Persist the customer data to Lambda local storage. Configure new Lambda functions to scan the local storage to save the customer data to the database.

Lambda local storage is ephemeral and is not guaranteed to persist across invocations. Relying on it for durable storage is unreliable and not suitable for this scenario.

- ✔ Store the customer data in an Amazon Simple Queue Service (Amazon SQS) FIFO queue. Create a new Lambda function that polls the queue and stores the customer data in the database.

Using an SQS FIFO queue provides a reliable and durable buffer for customer data during database upgrades. A separate Lambda function can then poll the queue and process the data once the database is available, ensuring no data loss.

Overall Explanation:

The most effective solution is to use an Amazon SQS FIFO queue to store the data during database upgrades. This decouples the data ingestion from the database's availability. A separate Lambda function can then process the queue and reliably store the data into the database once it is back online. RDS Proxy (Option A) doesn't store data. Increasing Lambda runtime (Option B) is not reliable for longer outages. Lambda local storage (Option C) is ephemeral.

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Question 31

Not Attempted

Security



A company has two AWS accounts: Production and Development. The company needs to push code changes in the Development account to the Production account. During the alpha phase, only two senior developers on the development team require access to the Production account. In the beta phase, more developers will need access to perform testing. Which solution will meet these requirements?

- ⊗ Create two policy documents by using the AWS Management Console in each account. Assign the policy to developers who need access.

This option doesn't establish secure cross-account access. Policies created within each account only manage permissions within that account, not across accounts.

- ⊗ Create an IAM role in the Development account. Grant the IAM role access to the Production account. Allow developers to assume the role.

It's best practice to create IAM roles in the account being accessed (Production). This approach centralizes permission management in the Production account.

- ✔ Create an IAM role in the Production account. Define a trust policy that specifies the Development account. Allow developers to assume the role.

This is the recommended approach. An IAM role in the Production account with a trust relationship to the Development account allows developers in the Development account to assume the role and gain temporary access to Production resources.

- ⊗ Create an IAM group in the Production account. Add the group as a principal in a trust policy that specifies the Production account. Add developers to the group.

IAM groups cannot be directly used as principals in trust policies for cross-account access. Trust policies define which accounts or entities are allowed to assume the role.

Overall Explanation:

Creating an IAM role in the Production account with a trust policy that specifies the Development account is the correct approach. This allows developers in the Development account to assume the role in Production and gain temporary access. In the alpha phase, the trust policy will allow only the two senior developers to assume the role. Later in the beta phase, more developers can be granted access through the role's trust policy. This approach facilitates secure, cross-account access while maintaining control over who has access to the Production environment. Groups can't be used in a trust policy. Option B violates AWS best practices by creating the IAM role in the development account.

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Question 32

Not Attempted

Security



A company wants to restrict access to the content of its web application. The company needs to protect the content by using authorization techniques that are available on AWS. The company also wants to implement a serverless architecture for authorization and authentication that has low login latency. The solution must integrate with the web application and serve web content globally. The application currently has a small user base, but the company expects the application's user base to increase. Which solution will meet these requirements?

- ✓ Configure Amazon Cognito for authentication. Implement Lambda@Edge for authorization. Configure Amazon CloudFront to serve the web application globally. This solution uses Cognito for authentication, providing a scalable and managed user directory. Lambda@Edge allows for authorization logic to be executed at the edge, reducing latency and improving performance globally via CloudFront.
- ✗ Configure AWS Directory Service for Microsoft Active Directory for authentication. Implement AWS Lambda for authorization. Use an Application Load Balancer to serve the web application globally. Using Active Directory adds unnecessary complexity and management overhead, and doesn't directly address the requirement for global content delivery. While Lambda can perform authorization, it won't be as performant as Lambda@Edge for global authorization and an ALB doesn't inherently provide global content delivery.
- ✗ Configure Amazon Cognito for authentication. Implement AWS Lambda for authorization. Use Amazon S3 Transfer Acceleration to serve the web application globally. S3 Transfer Acceleration improves upload speeds to S3, not global content delivery for a web application. While Cognito is suitable for authentication and Lambda can do authorization, S3 Transfer Acceleration is not correct here.
- ✗ Configure AWS Directory Service for Microsoft Active Directory for authentication. Implement Lambda@Edge for authorization. Use AWS Elastic Beanstalk to serve the web application globally. Active Directory adds unnecessary complexity and management overhead. Elastic Beanstalk provides a platform for deploying web applications but doesn't inherently provide global content delivery or the optimal architecture for it.

Overall Explanation:

The best solution is to use Amazon Cognito for authentication and Lambda@Edge for authorization with Amazon CloudFront serving the web application globally. Cognito provides a managed authentication service. Lambda@Edge allows you to execute authorization logic at the edge, closer to the user, providing low latency and enhanced security. CloudFront ensures the content is served globally and efficiently.

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A development team uses multiple AWS accounts for their development, staging, and production environments. Team members have been launching large Amazon EC2 instances that are often underutilized. A solutions architect needs to prevent the launching of large EC2 instances across all AWS accounts with the least operational overhead. What approach should the solutions architect take?

- ☐ Update the IAM policies to deny the launch of large EC2 instances. Apply the policies to all users in each account.

Updating IAM policies in each account requires separate management and maintenance, leading to increased operational overhead.

- ☐ Define a resource in AWS Resource Access Manager that prevents the launch of large EC2 instances.

AWS Resource Access Manager (RAM) is primarily for resource sharing and doesn't directly restrict resource launches, making it unsuitable for this scenario.

- ☐ Create an IAM role in each account that denies the launch of large EC2 instances. Grant the developers IAM group access to the role.

Creating IAM roles in each account adds to the operational overhead because these roles need to be managed and maintained across all accounts.

- ☒ Create an organization in AWS Organizations in the management account with the default policy. Create a service control policy (SCP) that denies the launch of large EC2 instances, and apply it to the AWS accounts.

SCPs are centrally managed within AWS Organizations and apply to all member accounts. They effectively prevent launching large instances with minimal operational overhead. SCPs override IAM permissions.

Overall Explanation:

Service Control Policies (SCPs) within AWS Organizations provide a centralized mechanism to manage permissions across multiple AWS accounts. By creating an SCP that denies the launch of large EC2 instances and applying it to the AWS accounts, the solutions architect can effectively prevent these instances from being launched with the least operational overhead. This approach avoids the need to manage individual IAM policies or roles in each account. AWS Resource Access Manager is for resource sharing and not preventing EC2 launches. IAM policies would require individual updates for each account.

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Question 34

Not Attempted

Monitoring & Management



A company has migrated a fleet of hundreds of on-premises virtual machines (VMs) to Amazon EC2 instances. The instances run a diverse fleet of Windows Server versions along with several Linux distributions. The company needs a solution that will automate the inventory and updating of the operating systems. The company also requires a summary of common vulnerabilities of each instance for regular monthly reviews. What should a solutions architect recommend to meet these requirements?

- ☐ Set up AWS Systems Manager Patch Manager to manage all the EC2 instances. Configure AWS Security Hub to produce monthly reports.

AWS Security Hub aggregates security alerts and performs security best practice checks but is not the primary tool for vulnerability scanning or patching.

- ☒ Set up AWS Systems Manager Patch Manager to manage all the EC2 instances. Deploy Amazon Inspector, and configure monthly reports.

AWS Systems Manager Patch Manager automates OS and application patching, and Amazon Inspector scans for vulnerabilities. This solution meets both requirements.

- ☐ Set up AWS Shield Advanced, and configure monthly reports. Deploy AWS Config to automate patch installations on the EC2 instances.

AWS Shield Advanced protects against DDoS attacks and AWS Config is primarily for configuration management and tracking changes, not patching. This does not meet the OS inventory and vulnerability scanning requirements.

- ☐ Set up Amazon GuardDuty in the account to monitor all EC2 instances. Deploy AWS Config to automate patch installations on the EC2 instances.

Amazon GuardDuty provides threat detection, while AWS Config focuses on configuration management. Neither fulfills the requirements for OS inventory, patch management, or vulnerability summaries.

Overall Explanation:

The correct solution is to use AWS Systems Manager Patch Manager for automated OS and application patching across the diverse fleet of EC2 instances. Additionally, Amazon Inspector should be deployed to scan the instances for vulnerabilities and generate

monthly reports summarizing common vulnerabilities. Therefore, option B provides the best solution for meeting the stated requirements.

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Question 35

Not Attempted

Databases



A company hosts its application in the AWS Cloud. The application runs on Amazon EC2 instances in an Auto Scaling group behind an Elastic Load Balancing (ELB) load balancer. The application connects to an Amazon DynamoDB table. For disaster recovery (DR) purposes, the company wants to ensure that the application is available from another AWS Region with minimal downtime. Which solution will meet these requirements with the LEAST downtime?

- ☒ Create an Auto Scaling group and an ELB in the DR Region. Configure the DynamoDB table as a global table. Configure DNS failover to point to the new DR Region's ELB.

This approach allows for quick failover since the infrastructure is already in place in the DR region and DynamoDB is already replicating data. DNS failover provides a fast and automated way to redirect traffic.

- ☐ Create an AWS CloudFormation template to create EC2 instances, ELBs, and DynamoDB tables to be launched when necessary. Configure DNS failover to point to the new DR Region's ELB.

Creating infrastructure using CloudFormation during a DR event will introduce significant downtime as it takes time to provision resources.

- ☐ Create an AWS CloudFormation template to create EC2 instances and an ELB to be launched when necessary. Configure the DynamoDB table as a global table. Configure DNS failover to point to the new DR Region's ELB.

Creating infrastructure using CloudFormation during a DR event will introduce significant downtime as it takes time to provision resources.

- ☐ Create an Auto Scaling group and an ELB in the DR Region. Configure the DynamoDB table as a global table. Create an Amazon CloudWatch alarm with an evaluation period of 10 minutes to invoke an AWS Lambda function that updates Amazon Route 53 to point to the DR Region's ELB.

While this approach pre-provisions infrastructure, using a CloudWatch alarm and Lambda function to update Route 53 adds unnecessary complexity and potential delay compared to direct DNS failover.

Overall Explanation:

The solution that provides the least downtime involves pre-provisioning the application infrastructure (Auto Scaling group and ELB) in the DR Region and configuring the DynamoDB table as a global table. Using DNS failover provides the quickest and most efficient way to redirect traffic to the DR Region in the event of a failure in the primary Region. CloudFormation based approaches will induce delay, and using Cloudwatch alarms with Lambda is more complex than required for direct DNS failover.

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Question 36

Not Attempted

Networking



A company runs an application on Amazon EC2 instances within a private subnet. The application requires storing and retrieving data in Amazon S3 buckets. To comply with regulatory requirements, the data must not traverse the public internet. Which solution should a solutions architect implement to satisfy these requirements most cost-effectively?

☐ Deploy a NAT gateway to access the S3 buckets.

A NAT gateway allows instances in a private subnet to connect to the internet or other AWS services, but it does not prevent traffic from traversing the public internet. This option is not cost-effective and does not meet the regulatory requirements.

☐ Deploy AWS Storage Gateway to access the S3 buckets.

AWS Storage Gateway is used to integrate on-premises environments with AWS storage services. While it can access S3, it's not the most direct or cost-effective solution for accessing S3 from EC2 instances within the same AWS region when avoiding the public internet.

☐ Deploy an S3 interface endpoint to access the S3 buckets.

An S3 interface endpoint creates a private network interface (ENI) in your VPC that connects to Amazon S3. Although it provides private access, it's generally more expensive due to the cost associated with the network interface and data processing charges compared to a gateway endpoint.

☒ Deploy an S3 gateway endpoint to access the S3 buckets.

An S3 gateway endpoint creates a route entry in the VPC route table for accessing S3 without crossing the public internet. Gateway endpoints are free and provide a cost-effective solution for accessing S3 from within a VPC.

- ⊗ Create an Amazon CloudWatch alarm. Configure the alarm to restart EC2 instances that transition to a stopped state.

While helpful, this only addresses instances that are stopped. It doesn't address instances that are unhealthy but still running, or instances that are impacted by an Availability Zone outage.

Overall Explanation:

The MOST cost-effective solution to achieve high availability is to combine a Network Load Balancer (NLB) with an Auto Scaling group configured across multiple Availability Zones. The NLB distributes traffic at the transport layer, while the Auto Scaling group ensures that healthy instances are available to handle the load, even if an Availability Zone fails. Option B provides the correct load balancing strategy and option D provides the necessary autoscaling and multi-AZ functionality.

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Question 38

Not Attempted

Serverless



A company hosts its static website on Amazon S3. They want to implement a contact form that allows users to submit their name, email, phone number, and a message. Given the website has fewer than 100 visits per month and the company needs to be notified via email for each submission, which solution offers the MOST cost-effective implementation?

- ⊗ Host the dynamic contact form in Amazon Elastic Container Service (Amazon ECS). Set up Amazon Simple Email Service (Amazon SES) to connect to a third-party email provider.

ECS may be overkill and more expensive for a low-traffic contact form. Additionally, using a third-party email provider adds complexity and potential cost.

- ✔ Create an Amazon API Gateway endpoint that invokes an AWS Lambda function to return the contact form. Configure another Lambda function on the API Gateway to publish a message to an Amazon Simple Notification Service (Amazon SNS) topic.

This solution leverages serverless technologies (API Gateway, Lambda, SNS), which are cost-effective for low-traffic applications. Lambda charges per invocation, making it ideal for infrequent submissions. SNS provides an easy way to notify the company via email.

- ⊗ Host the website by using AWS Amplify Hosting for static content and dynamic content. Use server-side scripting to build the contact form. Configure Amazon Simple Queue

Service (Amazon SQS) to deliver the message to the company.

While Amplify can host dynamic content, using SQS to deliver notifications might not be as straightforward or efficient as using SNS for email notifications.

- ⊗ Migrate the website from Amazon S3 to Amazon EC2 instances that run Windows Server. Use Internet Information Services (IIS) for Windows Server to host the webpage. Use client-side scripting to build the contact form. Integrate the form with Amazon WorkMail.

Migrating to EC2 is significantly more expensive than using S3 for static hosting and is unnecessary for a simple contact form. Client-side scripting with WorkMail integration may have limitations and be less reliable for server-side processing and notifications.

Overall Explanation:

The most cost-effective solution is to utilize serverless technologies like API Gateway, Lambda, and SNS. Option B allows for a dynamic contact form using Lambda, triggered by API Gateway. The SNS topic then provides a simple and reliable method for notifying the company via email about each submission. ECS and EC2 based solutions are less cost effective for small contact form needs.

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Question 39

Not Attempted

Security



A company uses AWS Organizations with dedicated AWS accounts for each business unit. Recently, a critical notification was mistakenly sent to the root user email address of a business unit account instead of the designated account owner. The company wants to ensure all future notifications are routed to the appropriate teams (billing, operations, or security) based on notification category. What is the MOST secure solution?

- ⊗ Configure each AWS account to use a single, company-managed email address accessible to all account owners. Configure alternate contacts for each account with corresponding distribution lists for the billing, security, and operations teams for each business unit.

This option requires all account owners to access a single email account, which is not ideal for security and separation of duties.

- ⊗ Configure each AWS account to use a different email distribution list for each business unit. Ensure each distribution list has administrator email addresses to respond to alerts. Configure alternate contacts for each AWS account with corresponding distribution lists for the billing team, the security team, and the operations team for each business unit.

While using distribution lists is a good approach, having admin email addresses to respond might not be the most secure way of directing alerts. There are better approaches to routing notifications by category.

- ⊗ Configure each AWS account root user email address to be the individual company-managed email address of one person from each business unit. Configure alternate contacts for each AWS account with corresponding distribution lists for the billing team, the security team, and the operations team for each business unit.

This option ties root user access to a single individual, potentially creating a single point of failure and security risk.

- ✔ Configure each AWS account root user to use email aliases that forward to a centralized mailbox. Configure alternate contacts for each account using a single, business-managed email distribution list for the billing, security, and operations teams respectively.

This approach uses email aliases for the root user, directing those notifications to a centralized location. This is combined with separate business-managed distribution lists for alternate contacts based on notification category (billing, operations, security), providing proper routing and improving security by not sending sensitive information directly to individuals.

Overall Explanation:

The most secure solution (Option D) involves using email aliases for root user notifications and utilizing dedicated, business-managed distribution lists for alternate contacts based on notification category. This centralizes control, reduces the risk of unauthorized access, and ensures appropriate routing of notifications to the relevant teams. Configuring email aliases for root users to a centralized mailbox avoids sending critical notifications to individual emails. Using distribution lists for alternate contacts allows routing the emails to the correct teams.

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A company runs an ecommerce application on AWS. Amazon EC2 instances process purchases and store the purchase details in an Amazon Aurora PostgreSQL DB cluster. Customers are experiencing application timeouts during times of peak usage. A solutions architect needs to rearchitect the application so that the application can scale to meet peak usage demands. Which combination of actions will meet these requirements MOST cost-effectively? (Choose two.)

- ☐ ⓧ Configure an Auto Scaling group of new EC2 instances to retry the purchases until the processing is complete. Update the applications to connect to the DB cluster by using Amazon RDS Proxy.

RDS Proxy is primarily for connection pooling and availability, addressing connection limits rather than performance bottlenecks related to processing purchases. Retrying purchases adds complexity and doesn't resolve the underlying scaling issue.

- ☐ ⓧ Configure the application to use an Amazon ElastiCache cluster in front of the Aurora PostgreSQL DB cluster.

ElastiCache is suitable for read-heavy workloads. However, purchase transactions primarily involve writes, making caching less effective in addressing the timeout issues during peak usage.

- ☒ ⓧ Update the application to send the purchase requests to an Amazon Simple Queue Service (Amazon SQS) queue. Configure an Auto Scaling group of new EC2 instances that read from the SQS queue.

Using SQS decouples the application and provides a buffer for purchase requests. An Auto Scaling group of EC2 instances consuming messages from the SQS queue allows the application to scale independently and handle bursts in traffic effectively.

- ☐ ⓧ Configure an AWS Lambda function to retry the ticket purchases until the processing is complete.

Lambda functions are not ideal for long-running database transactions and can become cost-prohibitive at scale due to invocation rates.

- ☐ ⓧ Configure an Amazon API Gateway REST API with a usage plan.

API Gateway provides API management capabilities, but does not directly address the scaling and timeout issues related to purchase processing and database interactions.

Overall Explanation:

Using SQS to queue purchase requests allows for decoupling and buffering. The Auto Scaling group of EC2 instances will then process these requests at their own pace, scaling

according to demand. This is more cost effective than options using Lambda for retries, or solely relying on caching or RDS proxy. C is correct.

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Question 41

Not Attempted

Storage



A survey company stores 3 TB of data in an Amazon S3 bucket, which is growing daily. They need to share this data with a marketing firm that also uses S3 buckets. To minimize the survey company's data transfer costs, which solution should they implement?

- ☒ **Configure the Requester Pays feature on the company's S3 bucket.**
With Requester Pays, the marketing firm bears the data transfer costs when accessing the data in the company's S3 bucket, minimizing costs for the survey company.
- ☐ **Configure S3 Cross-Region Replication from the company's S3 bucket to one of the marketing firm's S3 buckets.**
Cross-Region Replication would incur additional storage and data transfer costs for the survey company, increasing their expenses instead of reducing them.
- ☐ **Configure cross-account access for the marketing firm so that the marketing firm has access to the company's S3 bucket.**
While this allows access, it doesn't inherently address who pays for the data transfer. The survey company would still likely be responsible for data transfer costs.
- ☐ **Configure the company's S3 bucket to use S3 Intelligent-Tiering. Sync the S3 bucket to one of the marketing firm's S3 buckets.**
S3 Intelligent-Tiering optimizes storage costs but doesn't reduce data transfer costs when the marketing firm accesses the data. Syncing the buckets would also incur costs for the survey company.

Overall Explanation:

The most cost-effective solution is to configure Requester Pays on the survey company's S3 bucket. This ensures the marketing firm pays for data transfer costs, minimizing the survey company's expenses. Replication incurs costs, and simple cross-account access doesn't shift data transfer payment responsibility.

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Question 42

Not Attempted

Cost Optimization



A company uses AWS Organizations with 150 applications spread across 30 AWS accounts. They've configured an AWS Cost and Usage Report in the management account, delivering it to an Amazon S3 bucket which is replicated to another bucket in a data collection account. Senior leadership requires a custom dashboard displaying daily NAT gateway costs, starting from the current month's beginning. What solution fulfills these requirements?

- ☐ Share an Amazon QuickSight dashboard with a table visual, configuring QuickSight to use AWS DataSync to query the report.

DataSync is used for data transfer, not querying data in S3 for visualization.

- ☒ Share an Amazon QuickSight dashboard with a table visual, configuring QuickSight to use Amazon Athena to query the report.

This is the correct answer. QuickSight offers dashboarding capabilities for senior leadership, and Athena allows querying of the Cost and Usage Report data in S3.

- ☐ Share an Amazon CloudWatch dashboard with a table visual, configuring CloudWatch to use AWS DataSync to query the report.

CloudWatch is designed for monitoring metrics and logs and isn't the ideal solution for custom dashboards for senior leadership. DataSync is also not appropriate for querying the data.

- ☐ Share an Amazon CloudWatch dashboard with a table visual, configuring CloudWatch to use Amazon Athena to query the report.

While CloudWatch can display some metrics, it's not suited for displaying custom reports in a user-friendly format for leadership consumption, and the table visualization requirement would be hard to achieve. QuickSight is much better for dashboarding.

Overall Explanation:

The best solution is to use Amazon QuickSight for the dashboarding, given its strong visualization capabilities. To query the data in the S3 bucket containing the AWS Cost and Usage Report, Amazon Athena is the most appropriate choice. Athena allows you to use SQL to directly query data in S3. Therefore, the correct answer is B.

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Question 43

Not Attempted

Networking



A company is hosting a high-traffic static website on Amazon S3 with an Amazon CloudFront distribution that has a default TTL of 0 seconds. The company wants to implement caching to improve performance for the website. However, the company also wants to ensure that stale content is not served for more than a few minutes after a deployment. Which combination of caching methods should a solutions architect implement to meet these requirements? (Choose two.)

- ☒ Set the CloudFront default TTL to 2 minutes.
Setting the default TTL to 2 minutes allows CloudFront to cache content for a short period, ensuring content is cached and served quickly.
- ☐ Set a default TTL of 2 minutes on the S3 bucket.
TTL is not a concept applied directly to S3 buckets. TTL and caching behaviors are managed through Cache-Control headers and CloudFront settings.
- ☐ Add a Cache-Control private directive to the objects in Amazon S3.
The `private` directive lets caching occur locally only, not in shared caches like CloudFront, preventing shared caching.
- ☐ Create an AWS Lambda@Edge function to add an Expires header to HTTP responses. Configure the function to run on viewer response.
While Lambda@Edge offers fine-grained control, setting the default TTL in CloudFront is simpler and sufficient for the requirements. Plus, Lambda@Edge is not necessary to ensure content is served fresh after a deployment when using CloudFront invalidation.
- ☒ Add a Cache-Control max-age directive of 24 hours to the objects in Amazon S3. On deployment, create a CloudFront invalidation to clear any changed files from edge caches.
Using a long Cache-Control `max-age` directive improves performance, while targeted CloudFront invalidations ensure that updated content is served immediately after deployment.

Overall Explanation:

The optimal solution is to set the CloudFront default TTL to 2 minutes and add a Cache-Control `max-age` directive to objects in S3, coupled with CloudFront invalidations on

deployment. Setting a short TTL ensures that the browser asks CloudFront after 2 minutes. Adding `Cache-Control max-age` to S3 and invalidating on deployment ensures stale content is cleared and that updates are propagated efficiently. This combination balances performance and freshness of content.

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Question 44

Not Attempted

Serverless



A company's application uses Amazon EC2 instances and AWS Lambda functions. The EC2 instances reside in private subnets within a VPC. The Lambda functions require direct network access to the EC2 instances for proper operation. The application is intended to run for 1 year, and the number of Lambda functions utilized will increase throughout the year. The company aims to minimize costs across all application resources. Which solution best meets these requirements?

- ☐ Purchase an EC2 Instance Savings Plan. Connect the Lambda functions to the private subnets that contain the EC2 instances.

An EC2 Instance Savings Plan only applies to EC2 usage, and does not provide cost savings for Lambda functions. Therefore this is incorrect.

- ☐ Purchase an EC2 Instance Savings Plan. Connect the Lambda functions to new public subnets in the same VPC where the EC2 instances run.

An EC2 Instance Savings Plan only applies to EC2 usage. Lambda functions in public subnets cannot directly communicate with EC2 instances in private subnets, further complicating the solution.

- ☒ Purchase a Compute Savings Plan. Connect the Lambda functions to the private subnets that contain the EC2 instances.

A Compute Savings Plan provides discounts on both EC2 and Lambda usage, making it suitable for the evolving needs of the application. Connecting Lambda functions to private subnets allows direct and secure access to EC2 instances.

- ☐ Purchase a Compute Savings Plan. Keep the Lambda functions in the Lambda service VPC.

Keeping the Lambda functions in the Lambda service VPC would not allow for direct network access to the EC2 instances in the private subnets.

Overall Explanation:

The most cost-effective and functional solution is to purchase a Compute Savings Plan, which covers both EC2 and Lambda usage, and to connect the Lambda functions to the private subnets containing the EC2 instances. This ensures direct network access while minimizing costs and maximizing resource utilization. A Compute Savings Plan accounts for increased Lambda usage throughout the year. EC2 Instance Savings Plans do not apply to Lambda function costs.

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Question 45

Not Attempted

Cost Optimization



A company has deployed a multi-account strategy on AWS by using AWS Control Tower. The company has provided individual AWS accounts to each of its developers. The company wants to implement controls to limit AWS resource costs that the developers incur. Which solution will meet these requirements with the LEAST operational overhead?

- ☐ Instruct each developer to tag all their resources with a tag that has a key of CostCenter and a value of the developer's name. Use the required-tags AWS Config managed rule to check for the tag. Create an AWS Lambda function to terminate resources that do not have the tag. Configure AWS Cost Explorer to send a daily report to each developer to monitor their spending.

This approach requires manual tagging by developers, creating and maintaining Lambda functions, and relies on daily reports, leading to high operational overhead and reactive cost control.

- ☒ Use AWS Budgets to establish budgets for each developer account. Set up budget alerts for actual and forecast values to notify developers when they exceed or expect to exceed their assigned budget. Use AWS Budgets actions to apply a DenyAll policy to the developer's IAM role to prevent additional resources from being launched when the assigned budget is reached.

AWS Budgets allows setting budgets for each developer account and triggering actions based on budget breaches. This includes applying IAM policies to restrict resource creation, providing proactive cost control with minimal operational overhead. Note that the 'DenyAll' policy may not exist as is, but custom deny policies can be used.

- ☐ Use AWS Cost Explorer to monitor and report on costs for each developer account. Configure Cost Explorer to send a daily report to each developer to monitor their

spending. Use AWS Cost Anomaly Detection to detect anomalous spending and provide alerts.

Cost Explorer provides monitoring and reporting capabilities, and anomaly detection can identify unusual spending. However, this approach is reactive and does not prevent developers from incurring costs exceeding their limits.

- ⊗ Use AWS Service Catalog to allow developers to launch resources within a limited cost range. Create AWS Lambda functions in each AWS account to stop running resources at the end of each work day. Configure the Lambda functions to resume the resources at the start of each work day.

Service Catalog can help control resource types, and Lambda functions can stop/start resources. However, managing Service Catalog and Lambda functions across multiple accounts increases operational overhead. It also doesn't effectively limit costs as developers can still launch resources within the allowed catalog during the day.

Overall Explanation:

The most effective solution with the least operational overhead is to utilize AWS Budgets. It offers proactive cost control by establishing budgets, sending alerts, and triggering actions to restrict resource creation when budget limits are reached. While tagging and reporting are useful, they don't actively prevent excessive spending. Service Catalog and Lambda functions introduce significant management overhead. Therefore, AWS Budgets provides the best balance of cost control and ease of management.

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Question 46

Not Attempted

Security



A solutions architect is designing a three-tier web application with security as a high priority. The architecture consists of an internet-facing Application Load Balancer (ALB), a web tier on EC2 instances in private subnets, an application tier with business logic on EC2 instances in private subnets, and a database tier with Microsoft SQL Server on EC2 instances in private subnets. Which combination of security group configurations should the solutions architect implement to achieve this? (Choose three.)

- ✔ Configure the security group for the web tier to allow inbound HTTPS traffic from the security group for the ALB.

This allows the ALB to send traffic to the web tier over HTTPS. Security groups are stateful, so return traffic is automatically allowed.

- ⊗ Configure the security group for the web tier to allow outbound HTTPS traffic to 0.0.0.0/0.

This is not necessary. Security groups are stateful, so outbound rules are not required for established connections.

- ✔ Configure the security group for the database tier to allow inbound Microsoft SQL Server traffic from the security group for the application tier.

This allows the application tier to communicate with the database tier using the Microsoft SQL Server protocol. Security groups are stateful, so return traffic is automatically allowed.

- ⊗ Configure the security group for the database tier to allow outbound HTTPS traffic and Microsoft SQL Server traffic to the security group for the web tier.

This is incorrect. Security groups should only allow traffic from tiers that need to communicate directly with that tier. The database tier does not need to talk to the web tier directly.

- ✔ Configure the security group for the application tier to allow inbound HTTPS traffic from the security group for the web tier.

This allows the web tier to send traffic to the application tier over HTTPS. Security groups are stateful, so return traffic is automatically allowed.

- ⊗ Configure the security group for the application tier to allow outbound HTTPS traffic and Microsoft SQL Server traffic to the security group for the web tier.

This is not necessary. The Application tier initiates connection to the DB, not to the web tier. The return traffic for existing connections is already allowed because of stateful nature of security groups.

Overall Explanation:

To secure the three-tier web application, the following security group configurations are necessary: 1. Allow inbound HTTPS traffic to the web tier from the ALB security group. 2. Allow inbound HTTPS traffic to the application tier from the web tier security group. 3. Allow inbound Microsoft SQL Server traffic to the database tier from the application tier security group. Security groups are stateful, meaning that return traffic for established connections is automatically allowed. This simplifies the configuration and enhances security by limiting the allowed traffic to only what is necessary for the application to function.

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Question 47

Not Attempted

Cost Optimization



A company has released a new version of its production application. The company's workload uses Amazon EC2, AWS Lambda, AWS Fargate, and Amazon SageMaker. The company wants to cost optimize the workload now that usage is at a steady state. The company wants to cover the most services with the fewest savings plans. Which combination of savings plans will meet these requirements? (Choose two.)

- ☐ Purchase an EC2 Instance Savings Plan for Amazon EC2 and SageMaker.
An EC2 Instance Savings Plan only applies to EC2 usage in a specific Region.
- ☐ Purchase a Compute Savings Plan for Amazon EC2, Lambda, and SageMaker.
A Compute Savings Plan covers EC2, Lambda, and Fargate, but not SageMaker.
- ☒ Purchase a SageMaker Savings Plan.
A SageMaker Savings Plan specifically covers SageMaker usage.
- ☒ Purchase a Compute Savings Plan for Lambda, Fargate, and Amazon EC2.
A Compute Savings Plan covers EC2, Lambda, and Fargate.
- ☐ Purchase an EC2 Instance Savings Plan for Amazon EC2 and Fargate.
An EC2 Instance Savings Plan only applies to EC2 usage in a specific Region.

Overall Explanation:

The best solution to cover all services with the fewest savings plans is to purchase a SageMaker Savings Plan and a Compute Savings Plan. A SageMaker Savings Plan covers the Amazon SageMaker costs. A Compute Savings Plan covers the AWS Lambda, AWS Fargate, and Amazon EC2 costs. Using this combination of savings plans maximizes the cost savings and minimizes the number of plans.

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Question 48

Not Attempted

Databases



A company's applications are connected to a Microsoft SQL Server database. The company plans to migrate to an Amazon Aurora PostgreSQL database while minimizing application code changes. Which combination of actions will meet these requirements? (Choose two.)

- ☐ Use the AWS Schema Conversion Tool (AWS SCT) to rewrite the SQL queries in the applications.

AWS SCT helps in converting the database schema, but doesn't directly rewrite SQL queries within the application code itself.

- ☒ Enable Babelfish on Aurora PostgreSQL to run the SQL queries from the applications.

Babelfish for Aurora PostgreSQL allows Aurora to understand commands from applications written for Microsoft SQL Server, minimizing the need to rewrite queries.

- ☒ Migrate the database schema and data by using the AWS Schema Conversion Tool (AWS SCT) and AWS Database Migration Service (AWS DMS).

AWS SCT handles schema conversion, and AWS DMS handles data migration. This is a standard approach for migrating databases.

- ☐ Use Amazon RDS Proxy to connect the applications to Aurora PostgreSQL.

Amazon RDS Proxy helps improve database scalability and availability, but it doesn't address the compatibility issues between SQL Server and Aurora PostgreSQL.

- ☐ Use AWS Database Migration Service (AWS DMS) to rewrite the SQL queries in the applications.

AWS DMS is used for data migration, not for rewriting SQL queries within the applications.

Overall Explanation:

Using AWS SCT and AWS DMS for migrating the database schema and data (Option C), coupled with enabling Babelfish on Aurora PostgreSQL to understand SQL Server commands (Option B), minimizes the need for application code changes during the migration. AWS SCT converts the schema, DMS migrates the data, and Babelfish allows the application to continue using familiar SQL Server syntax.

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A company plans to rehost an application to Amazon EC2 instances that use Amazon Elastic Block Store (Amazon EBS) as the attached storage. A solutions architect must design a solution to ensure that all newly created Amazon EBS volumes are encrypted by default. The solution must also prevent the creation of unencrypted EBS volumes. Which solution will meet these requirements?

- ☐ ⓧ Configure the EC2 account attributes to always encrypt new EBS volumes.

Configuring the EC2 account attributes will encrypt all newly created volumes. However, it doesn't prevent the creation of unencrypted volumes during instance launch if the request specifies no encryption. While this enables default encryption, it does not explicitly *prevent* unencrypted volume creation.

- ☒ ⓧ Use AWS Config. Configure the encrypted-volumes identifier. Apply the default AWS Key Management Service (AWS KMS) key.

AWS Config can be used to identify non-compliant EBS volumes and trigger remediation actions. By using AWS Config, you can enforce the rule that all EBS volumes must be encrypted and prevent the creation of unencrypted EBS volumes. This aligns with the requirement to prevent the creation of unencrypted volumes.

- ☐ ⓧ Configure AWS Systems Manager to create encrypted copies of the EBS volumes. Reconfigure the EC2 instances to use the encrypted volumes.

This solution involves creating encrypted copies of EBS volumes *after* they are created. This does not prevent the initial creation of unencrypted volumes and adds unnecessary complexity.

- ☐ ⓧ Create a customer managed key in AWS Key Management Service (AWS KMS). Configure AWS Migration Hub to use the key when the company migrates workloads.

AWS Migration Hub is used for migration tracking. Configuring it with a KMS key doesn't ensure default encryption of EBS volumes or prevent the creation of unencrypted EBS volumes. Migration Hub is not directly involved in EBS volume creation or encryption enforcement.

Overall Explanation:

The best solution is to use AWS Config to ensure all EBS volumes are encrypted. AWS Config can identify non-compliant resources, like unencrypted volumes, and automatically trigger remediation actions. Option A configures default encryption, but does not prevent unencrypted volume creation. Options C and D do not meet the requirements. Therefore, using AWS Config is the most comprehensive solution that prevents the creation of unencrypted EBS volumes.

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Question 50

Not Attempted

Serverless



An e-commerce company requires a solution to collect user clickstream data from its website for real-time analytics. The website experiences fluctuating traffic patterns throughout the day. The company needs a scalable solution that can dynamically adjust to these varying levels of traffic. Which of the following solutions best meets these requirements?

- ☒ Use a data stream in Amazon Kinesis Data Streams in on-demand mode to capture the clickstream data. Use AWS Lambda to process the data in real time.
This is the best solution because Kinesis Data Streams in on-demand mode automatically scales based on traffic, making it ideal for fluctuating traffic patterns. AWS Lambda enables real-time processing of the data without managing infrastructure.
- ☐ Use Amazon Kinesis Data Firehose to capture the clickstream data. Use AWS Glue to process the data in real time.
While Kinesis Data Firehose is scalable, it's geared more towards near real-time delivery, not strictly real-time. AWS Glue is better suited for ETL and batch processing, not real-time analysis.
- ☐ Use Amazon Kinesis Video Streams to capture the clickstream data. Use AWS Glue to process the data in real time.
Kinesis Video Streams is designed specifically for video and audio streaming data, not clickstream data. AWS Glue is better suited for ETL and batch processing, not real-time analysis.
- ☐ Use Amazon Managed Service for Apache Flink (previously known as Amazon Kinesis Data Analytics) to capture the clickstream data. Use AWS Lambda to process the data in real time.
While Flink can be used for stream processing, it's not typically used to initially capture and ingest the raw clickstream data directly. Kinesis Data Streams is more suitable for initial data capture.

Overall Explanation:

The optimal solution is to use Amazon Kinesis Data Streams in on-demand mode to capture clickstream data and AWS Lambda to process it in real time. Kinesis Data Streams in on-demand mode provides the necessary scalability to handle fluctuating traffic, and AWS Lambda allows for real-time data processing without infrastructure management.

Kinesis Data Firehose is more for near real-time delivery, Kinesis Video Streams is for video, and Kinesis Data Analytics isn't the best service for capturing the initial stream.

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Question 51

Not Attempted

Storage



A global company runs its workloads on AWS. The company's application uses Amazon S3 buckets across AWS Regions for sensitive data storage and analysis. The company stores millions of objects in multiple S3 buckets daily. The company wants to identify all S3 buckets that are *not* versioning-enabled across all regions. Which solution will meet these requirements?

- ☐ Set up an AWS CloudTrail event that has a rule to identify all S3 buckets that are not versioning-enabled across Regions.
CloudTrail primarily logs API calls made to AWS services. While CloudTrail can track changes to S3 buckets, it doesn't provide a built-in mechanism to efficiently identify buckets without versioning enabled across all regions.
- ☒ Use Amazon S3 Storage Lens to identify all S3 buckets that are not versioning-enabled across Regions.
S3 Storage Lens provides organization-wide visibility into object storage, including metrics on data protection best practices like versioning. It can directly identify buckets where versioning is not enabled.
- ☐ Enable IAM Access Analyzer for S3 to identify all S3 buckets that are not versioning-enabled across Regions.
IAM Access Analyzer for S3 helps identify buckets with access that is shared with external entities. It does not directly provide metrics related to S3 versioning status.
- ☐ Create an S3 Multi-Region Access Point to identify all S3 buckets that are not versioning-enabled across Regions.
S3 Multi-Region Access Points are used to simplify access to data in multiple regions but do not offer any insights or functionality related to versioning status across buckets.

Overall Explanation:

Amazon S3 Storage Lens is the best solution because it is specifically designed to provide visibility into object storage usage and data protection best practices, including identifying S3 buckets that are not versioning-enabled across different regions. It offers metrics like 'Versioning-enabled bucket count' to easily find buckets without versioning. CloudTrail can't directly identify buckets without versioning enabled. IAM Access Analyzer focuses on access control, not versioning. S3 Multi-Region Access Points simplifies data access but doesn't provide versioning information.

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Question 52

Not Attempted

Security



A company stores confidential audit documents in an Amazon S3 bucket. The S3 bucket policy restricts access to audit team IAM user credentials following the principle of least privilege. Company managers are concerned about accidental deletion of documents in the S3 bucket and desire a more secure solution. What should a solutions architect implement to secure the audit documents?

- ☒ Enable the versioning and MFA Delete features on the S3 bucket.
Enabling versioning allows recovery of accidentally deleted objects, while MFA Delete requires multi-factor authentication for deletion, preventing accidental or unauthorized permanent deletions.
- ☐ Enable multi-factor authentication (MFA) on the IAM user credentials for each audit team IAM user account.
While MFA enhances user authentication security, it doesn't directly prevent accidental object deletion in the S3 bucket.
- ☐ Add an S3 Lifecycle policy to the audit team's IAM user accounts to deny the s3:DeleteObject action during audit dates.
Lifecycle policies can manage object lifecycles but using them to deny deletion only during audit dates doesn't protect against accidental deletions at other times.
- ☐ Use AWS Key Management Service (AWS KMS) to encrypt the S3 bucket and restrict audit team IAM user accounts from accessing the KMS key.
KMS encryption secures data at rest, but doesn't prevent accidental object deletion. Restricting KMS key access would prevent access to the encrypted objects but wouldn't prevent deletion if the IAM user had the s3:DeleteObject permission.

Overall Explanation:

Enabling versioning allows for the recovery of accidentally deleted objects. Enabling MFA Delete provides an additional layer of security by requiring multi-factor authentication to permanently delete an object, mitigating the risk of accidental deletion.

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Question 53

Not Attempted

Storage



A company needs to optimize its Amazon S3 storage costs for an application that generates many files that cannot be recreated. Each file is approximately 5 MB and is stored in Amazon S3 Standard storage. The company must store the files for 4 years before the files can be deleted. The files must be immediately accessible. The files are frequently accessed in the first 30 days of object creation, but they are rarely accessed after the first 30 days. Which solution will meet these requirements MOST cost-effectively?

- ☐ Create an S3 Lifecycle policy to move the files to S3 Glacier Instant Retrieval 30 days after object creation. Delete the files 4 years after object creation.
While Glacier Instant Retrieval offers immediate access, it has a minimum storage duration of 90 days and storage costs may be higher than S3 Standard-IA when access is infrequent after 30 days.
- ☐ Create an S3 Lifecycle policy to move the files to S3 One Zone-Infrequent Access (S3 One Zone-IA) 30 days after object creation. Delete the files 4 years after object creation.
S3 One Zone-IA is cheaper but stores data in a single Availability Zone. Because the files cannot be recreated, this option violates the durability requirement.
- ☒ Create an S3 Lifecycle policy to move the files to S3 Standard-Infrequent Access (S3 Standard-IA) 30 days after object creation. Delete the files 4 years after object creation.
S3 Standard-IA is optimized for objects accessed less frequently but still requires rapid retrieval and high durability. This option provides immediate access and cost savings after the initial 30 days while maintaining data durability.
- ☐ Create an S3 Lifecycle policy to move the files to S3 Standard-Infrequent Access (S3 Standard-IA) 30 days after object creation. Move the files to S3 Glacier Flexible Retrieval 4 years after object creation.
Moving the files to Glacier after 4 years is unnecessary, as the company needs to delete the files after 4 years to meet the storage requirement. Deleting the files directly from S3 Standard-IA after 4

years provides the most cost-effective solution.

Overall Explanation:

S3 Standard-IA offers the best balance of cost and accessibility for infrequently accessed data that still requires rapid retrieval and high durability. Moving the files to S3 Standard-IA after the first 30 days allows the company to leverage the lower storage costs while maintaining immediate access and deleting them after 4 years. Therefore, the files are stored in S3 Standard-IA storage class for the remaining duration which reduces the costs. S3 One Zone-IA is not suitable as data can not be recreated and Glacier is not required as data is deleted after 4 years.

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Question 54

Not Attempted

Storage



A company operates a critical storage application in the AWS Cloud, utilizing Amazon S3 in two AWS Regions. The application needs to direct remote user data to the S3 bucket nearest to them, avoiding public network congestion. The company also requires the application to automatically failover between regions with minimal S3 management overhead. Which solution best meets these requirements?

- ⊗ Implement an active-active design between the two Regions. Configure the application to use the regional S3 endpoints closest to the user.

While this uses both regions, it doesn't leverage a global endpoint and requires the application to manage routing, potentially increasing complexity and not providing seamless failover.

- ⊗ Use an active-passive configuration with S3 Multi-Region Access Points. Create a global endpoint for each of the Regions.

Active-passive doesn't fully utilize both regions for performance and cost optimization. Also, creating multiple global endpoints defeats the purpose of a centralized access point.

- ⊗ Send user data to the regional S3 endpoints closest to the user. Configure an S3 cross-account replication rule to keep the S3 buckets synchronized.

This approach necessitates managing the regional endpoint selection within the application and does not inherently provide failover capabilities without additional logic. It also doesn't guarantee data is sent via the AWS network, thus not fulfilling the 'no public network congestion' requirement.

- ✓ Set up Amazon S3 to use Multi-Region Access Points in an active-active configuration with a single global endpoint. Configure S3 Cross-Region Replication.

This is the best solution. Multi-Region Access Points in active-active configuration with S3 Cross-Region Replication directs requests to the nearest region and provides automatic failover. The global endpoint simplifies application integration. S3 Cross-Region Replication ensures data consistency across regions.

Overall Explanation:

Option D is the most suitable because it leverages Amazon S3 Multi-Region Access Points in an active-active configuration, directing requests to the closest AWS Region with low latency via the AWS global network. The single global endpoint simplifies application connectivity and the application does not need to implement failover logic. S3 Cross-Region Replication ensures data consistency across regions, enabling seamless failover capabilities in case of an outage in one region.

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Question 55

Not Attempted

Compute



A company is migrating a data center from its on-premises location to AWS. The company has several legacy applications that are hosted on individual virtual servers, and changes to the application designs cannot be made. Each virtual server currently runs as its own EC2 instance. A solutions architect needs to ensure that the applications are reliable and fault tolerant after migration to AWS. The applications will run on Amazon EC2 instances. Which solution will meet these requirements?

- ✓ Create an Auto Scaling group that has a minimum of one and a maximum of one. Create an Amazon Machine Image (AMI) of each application instance. Use the AMI to create EC2 instances in the Auto Scaling group. Configure an Application Load Balancer in front of the Auto Scaling group.

An Auto Scaling group with min/max of 1 ensures one instance is always running, replacing it if it fails. Using an AMI ensures consistent configuration. An ALB manages traffic and prepares for potential scaling.

- ✗ Use AWS Backup to create an hourly backup of the EC2 instance that hosts each application. Store the backup in Amazon S3 in a separate Availability Zone. Configure a

disaster recovery process to restore the EC2 instance for each application from its most recent backup.

Hourly backups provide disaster recovery, but not real-time fault tolerance. The recovery process introduces downtime, failing to meet reliability requirements.

- ⊗ Create an Amazon Machine Image (AMI) of each application instance. Launch two new EC2 instances from the AMI. Place each EC2 instance in a separate Availability Zone. Configure a Network Load Balancer that has the EC2 instances as targets.

This option provides high availability but relies on manual intervention if an instance fails. Without Auto Scaling, new instances must be launched manually.

- ⊗ Use AWS Migration Hub Refactor Spaces to migrate each application off the EC2 instance. Break down functionality from each application into individual components. Host each application on Amazon Elastic Container Service (Amazon ECS) with an AWS Fargate launch type.

This option requires changing the application design, which is explicitly prohibited in the scenario.

Overall Explanation:

Option A provides fault tolerance within the given constraints. Auto Scaling ensures a replacement instance is launched automatically in case of failure, leveraging an AMI for consistent configuration. Although the min/max is one, Auto Scaling will still replace the failed instance. The ALB provides a stable endpoint and is forward-looking in case scaling becomes an option. Options B, C, and D all have drawbacks relating to downtime or a violation of the problem requirements.

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Question 56

Not Attempted

Networking



A company wants to isolate its workloads by creating an AWS account for each workload. The company needs a solution that centrally manages networking components for the workloads. The solution also must create accounts with automatic security controls (guardrails). Which solution will meet these requirements with the LEAST operational overhead?

- ✔ Use AWS Control Tower to deploy accounts. Create a networking account that has a VPC with private subnets and public subnets. Use AWS Resource Access Manager (AWS RAM) to share the subnets with the workload accounts.

This is the correct answer because AWS Control Tower simplifies account creation with built-in guardrails. Centralizing networking in a dedicated account and sharing subnets with AWS RAM minimizes operational overhead. AWS Control Tower also helps in automating account deployment and management. Control Tower is also designed to simplify the process of creating and managing a multi-account AWS environment while maintaining security and compliance.

- ⊗ Use AWS Organizations to deploy accounts. Create a networking account that has a VPC with private subnets and public subnets. Use AWS Resource Access Manager (AWS RAM) to share the subnets with the workload accounts.

While AWS Organizations helps in managing accounts, it does not provide the automatic security controls (guardrails) that AWS Control Tower offers. Therefore, it requires more operational overhead to implement and manage security controls.

- ⊗ Use AWS Control Tower to deploy accounts. Deploy a VPC in each workload account. Configure each VPC to route through an inspection VPC by using a transit gateway attachment.

This approach, while using Control Tower, introduces higher operational overhead due to the complexity of managing multiple VPCs and transit gateway attachments. Also, it is less efficient than centralizing networking components.

- ⊗ Use AWS Organizations to deploy accounts. Deploy a VPC in each workload account. Configure each VPC to route through an inspection VPC by using a transit gateway attachment.

This option has the highest operational overhead. It requires managing multiple VPCs, transit gateway attachments and lacks the automatic security control offered by AWS Control Tower.

Overall Explanation:

AWS Control Tower automates account deployment with built-in guardrails, addressing the need for automatic security controls. Centralizing networking in a dedicated account and using AWS RAM to share subnets efficiently manages networking components with minimal operational overhead. Therefore, using Control Tower and AWS RAM offers the least operational overhead while meeting the requirements.

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Question 57

Not Attempted

Cost Optimization



A company hosts a website on Amazon EC2 instances behind an Application Load Balancer (ALB). The website serves static content. Website traffic is increasing. The company wants to minimize the website hosting costs. Which solution will meet these requirements?

- ☒ Move the website to an Amazon S3 bucket. Configure an Amazon CloudFront distribution for the S3 bucket.

Moving static content to S3 and using CloudFront offers cost-effective scalability and content delivery, reducing the load on EC2 instances and minimizing hosting costs. CloudFront caches content, reducing latency and improving user experience.

- ☐ Move the website to an Amazon S3 bucket. Configure an Amazon ElastiCache cluster for the S3 bucket.

ElastiCache is primarily used for caching dynamic content or database queries. While S3 is suitable for static content, ElastiCache is not the appropriate service to handle increased traffic for static website content. ElastiCache isn't typically used in conjunction with S3 in this way, so this solution wouldn't be optimal or cost-effective.

- ☐ Move the website to AWS Amplify. Configure an ALB to resolve to the Amplify website.

AWS Amplify is suitable for hosting static websites, but using an ALB in front of it is unnecessary and adds complexity and cost. Amplify is designed to directly serve static content.

- ☐ Move the website to AWS Amplify. Configure EC2 instances to cache the website.

Using EC2 instances to cache a website hosted on Amplify defeats the purpose of using Amplify for static content hosting. It introduces unnecessary complexity and cost. Amplify is designed to directly serve static content efficiently.

Overall Explanation:

The most cost-effective solution for serving static content is to utilize Amazon S3 for storage and Amazon CloudFront for content delivery. This combination allows for scalable and efficient distribution of static assets, minimizing the load on EC2 instances and reducing overall hosting costs. Options B, C, and D introduce unnecessary services or configurations that increase costs and complexity.

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Question 58

Not Attempted

Storage



A company is migrating its on-premises file server to AWS. The company requires a solution that allows users to access files from their existing SMB clients. The solution should also minimize administrative overhead. Which AWS service should the company use?

- ☐ AWS Storage Gateway Volume Gateway with a created file share using the required client protocol and connected application server.
Volume Gateway does not directly support SMB. SMB is more commonly supported in File Gateway or FSx for Windows.
- ☐ AWS Storage Gateway Tape Gateway configured to use Amazon S3 with connected application server.
Tape Gateway is designed for backup and archival purposes, not for real-time file sharing using SMB.
- ☐ An Amazon EC2 Windows instance with a Windows file share role installed and configured, and connected application server.
While possible, this option involves manual setup, patching, maintenance, backups, and scaling, leading to significant administrative overhead.
- ☒ Amazon FSx for Windows File Server file system with connected application server.
Amazon FSx for Windows File Server is a fully managed service that provides native Windows file system capabilities, including SMB support, with minimal administrative overhead.

Overall Explanation:

Amazon FSx for Windows File Server is the best choice because it offers native SMB support with minimal administrative overhead. It is a fully managed service, simplifying setup, maintenance, and scaling compared to creating your own file server on EC2 or using Storage Gateway Volume Gateway. Tape Gateway is not suitable for this use case.

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Question 59

Not Attempted

[Databases](#)

A company is designing its production application's disaster recovery (DR) strategy. The application is backed by a MySQL database on an Amazon Aurora cluster in the us-east-1 Region. The company has chosen the us-west-1 Region as its DR Region. The company's target recovery point objective (RPO) is 5 minutes and the target recovery time objective (RTO) is 20

minutes. The company wants to minimize configuration changes. Which solution will meet these requirements with the MOST operational efficiency?

- ☐ Create an Aurora read replica in us-west-1 similar in size to the production application's Aurora MySQL cluster writer instance.

Read replicas do not automatically provide failover in the event of a disaster, and manual intervention could exceed the RTO.

- ☒ Convert the Aurora cluster to an Aurora global database. Configure managed failover.

Aurora Global Databases are designed for cross-region replication with minimal impact on database performance and enable fast failover, making it suitable for meeting the specified RPO and RTO with minimal operational overhead.

- ☐ Create a new Aurora cluster in us-west-1 that has Cross-Region Replication.

While Cross-Region Replication can replicate data, it doesn't automatically fail over. Manual intervention is required which exceeds RTO.

- ☐ Create a new Aurora cluster in us-west-1. Use AWS Database Migration Service (AWS DMS) to sync both clusters.

Using AWS DMS to sync clusters introduces complexity and may not meet the required RPO due to replication latency. Also it requires much more management overhead.

Overall Explanation:

Aurora Global Database provides a recovery point objective (RPO) of 1 second and a recovery time objective (RTO) of less than 1 minute, providing a strong foundation for a global business continuity plan. It allows for managed failover which eliminates the need for manual interventions.

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Question 60

Not Attempted

Compute



A company needs to run a critical, stateful data analysis job every week before the start of the work week. The job must run uninterrupted for at least 1 hour. What is the MOST suitable AWS solution for this scenario?

- ✓ Create a container for the job. Schedule the job to run as an AWS Fargate task on an Amazon Elastic Container Service (Amazon ECS) cluster by using Amazon EventBridge Scheduler.

This is the most suitable solution. Fargate allows for long-running, uninterrupted tasks, and EventBridge Scheduler provides reliable scheduling.

- ✗ Configure the job to run in an AWS Lambda function. Create a scheduled rule in Amazon EventBridge to invoke the Lambda function.

This is not a suitable solution. AWS Lambda functions have a maximum execution time of 15 minutes, which is insufficient for the 1-hour job duration.

- ✗ Configure an Auto Scaling group of Amazon EC2 Spot Instances that run Amazon Linux. Configure a crontab entry on the instances to run the analysis.

This is not a suitable solution. Spot Instances can be interrupted, violating the requirement for uninterrupted job execution.

- ✗ Configure an AWS DataSync task to run the job. Configure a cron expression to run the task on a schedule.

This is not a suitable solution. AWS DataSync is designed for data transfer, not for running long running analysis jobs.

Overall Explanation:

The best solution is to containerize the data analysis job and run it as a Fargate task scheduled by EventBridge Scheduler. Fargate offers a serverless compute environment suitable for long-running tasks, and EventBridge provides a reliable scheduling mechanism. Lambda functions are not suitable due to their time constraints, Spot Instances are prone to interruption, and DataSync is not designed for running applications.

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Question 61

Not Attempted

Security



A company runs workloads in the AWS Cloud. The company wants to centrally collect security data to assess security across the entire company and to improve workload protection. Which solution will meet these requirements with the LEAST development effort?

- ⊗ Configure a data lake in AWS Lake Formation. Use AWS Glue crawlers to ingest the security data into the data lake.

This option requires significant development effort to configure and manage AWS Lake Formation and Glue crawlers.

- ⊗ Configure an AWS Lambda function to collect the security data in .csv format. Upload the data to an Amazon S3 bucket.

This option requires development effort to create and maintain a Lambda function and manage the data in S3.

- ✓ Configure a data lake in Amazon Security Lake to collect the security data. Upload the data to an Amazon S3 bucket.

Amazon Security Lake is purpose-built for collecting and centralizing security data with minimal configuration and integrates easily with AWS security services. This option provides the least development effort.

- ⊗ Configure an AWS Database Migration Service (AWS DMS) replication instance to load the security data into an Amazon RDS cluster.

AWS DMS is not the correct service to load security data, as it is meant to migrate databases.

Overall Explanation:

Amazon Security Lake automatically centralizes security data from AWS environments, SaaS providers, on premises, and cloud sources into a purpose-built data lake, requiring the least development effort. It integrates easily with AWS security services, providing a streamlined solution with less manual configuration compared to using AWS Glue or Lambda. It is also designed to scale automatically, making it the most efficient and effective solution.

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Question 62

Not Attempted

Networking



A company is migrating five on-premises applications to VPCs in the AWS Cloud. Each application is currently deployed in isolated virtual networks on premises and should be deployed similarly in the AWS Cloud. The applications need to reach a shared services VPC. All the applications must be able to communicate with each other. If the migration is successful,

the company will repeat the migration process for more than 100 applications. Which solution will meet these requirements with the LEAST administrative overhead?

- ⊗ Deploy software VPN tunnels between the application VPCs and the shared services VPC. Add routes between the application VPCs in their subnets to the shared services VPC.

Deploying software VPN tunnels is complex, has high administrative overhead, and doesn't scale well to a large number of VPCs. It requires setting up and maintaining individual tunnels for each VPC.
- ⊗ Deploy VPC peering connections between the application VPCs and the shared services VPC. Add routes between the application VPCs in their subnets to the shared services VPC through the peering connection.

VPC peering is better than VPN tunnels but becomes complex and unmanageable as the number of VPCs increases. It requires manual peering for each pair of VPCs and routing configurations.
- ⊗ Deploy an AWS Direct Connect connection between the application VPCs and the shared services VPC. Add routes from the application VPCs in their subnets to the shared services VPC and the applications VPCs. Add routes from the shared services VPC subnets to the applications VPCs.

Direct Connect is costly and is designed for hybrid cloud setups or data center connectivity, not for inter-VPC communication within AWS.
- ✔ Deploy a transit gateway with associations between the transit gateway and the application VPCs and the shared services VPC. Add routes between the application VPCs in their subnets and the application VPCs to the shared services VPC through the transit gateway.

A transit gateway provides a centralized hub for managing inter-VPC communication. It simplifies the process of routing between VPCs, is highly scalable, and reduces the complexity of adding new VPCs, making it optimal for managing large-scale environments.

Overall Explanation:

The best solution is to deploy a transit gateway. It allows for inter-VPC communication for all applications/VPCs, can reach a shared services VPC, and easily scales for future migrations. VPN tunnels and VPC peering are not as scalable. Direct Connect is not appropriate for connecting numerous VPCs in the AWS Cloud.

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Question 63

Not Attempted

Storage



A company hosts an SMB file server in its data center. This server houses large files that are actively accessed for the first week after creation, after which access becomes infrequent. The data volume is nearing the storage capacity. To address this, a solutions architect needs to expand storage without compromising low-latency access to recent files and implement lifecycle management. Which solution satisfies these requirements?

- ☐ Use AWS DataSync to migrate data older than 7 days from the SMB file server to AWS. AWS DataSync focuses on data synchronization, not on providing extended storage and tiered access based on file age. It doesn't address the low-latency requirement for recent files.
- ☒ Implement an Amazon S3 File Gateway to extend the company's storage. Create an S3 Lifecycle policy to move data to S3 Glacier Deep Archive after 7 days. S3 File Gateway provides a local cache for frequently accessed files, ensuring low latency. The lifecycle policy automatically transitions older, less frequently accessed files to Glacier Deep Archive for cost-effective storage.
- ☐ Create an Amazon FSx for Windows File Server file system to extend the company's storage space. While FSx provides a managed Windows file server, it doesn't inherently provide file lifecycle management or tiering to a lower-cost storage solution like Glacier.
- ☐ Install a utility on each user's computer to access Amazon S3. Create an S3 Lifecycle policy to transition the data to S3 Glacier Flexible Retrieval after 7 days. Installing a utility on each user's computer is not a scalable solution and adds complexity. The direct user access to S3 also bypasses the local caching mechanism offered by S3 File Gateway, which is important for low-latency access to recently accessed files. Flexible retrieval isn't needed because the access is infrequent after 7 days.

Overall Explanation:

The best solution is to use Amazon S3 File Gateway to extend the company's storage with S3. The gateway caches recently accessed files for low-latency access. An S3 lifecycle policy then automatically transitions files to Glacier Deep Archive after 7 days to optimize costs for infrequently accessed data.

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Question 64

Not Attempted

Databases



A company has a publicly accessible SQL database containing movie data. The database runs on an Amazon RDS Single-AZ DB instance. A script executes queries at random intervals daily to record the number of new movies added to the database and reports a final total during business hours. The development team is experiencing inadequate database performance for development tasks while the script runs. A solutions architect needs to recommend a solution to resolve this with the LEAST operational overhead. Which solution will meet this requirement?

- ☐ **Modify the DB instance to be a Multi-AZ deployment.**
A Multi-AZ deployment improves high availability and fault tolerance, but it does not directly address the performance issue caused by the script's read queries.
- ☒ **Create a read replica of the database. Configure the script to query only the read replica.**
Creating a read replica and directing the script's queries to it offloads the read traffic from the primary database, resolving the performance issue with minimal operational overhead.
- ☐ **Instruct the development team to manually export the entries in the database at the end of each day.**
Manually exporting data introduces significant operational overhead and is not a scalable or efficient solution.
- ☐ **Use Amazon ElastiCache to cache the common queries that the script runs against the database.**
ElastiCache is useful for caching frequently accessed data. Since the script is looking for newly added movies, the data is constantly changing, making ElastiCache less effective and requiring more operational overhead to implement and manage.

Overall Explanation:

The correct solution is to create a read replica (Option 2). This allows the reporting script to run its queries against the read replica, isolating the workload and preventing it from impacting the performance of the primary database used for development. It requires minimal changes and offers the least operational overhead compared to the other options. Multi-AZ improves availability, manual exports are inefficient, and ElastiCache is not suitable for frequently changing data.

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